

Airframe Technology Master

Fatigue and Damage Tolerance of metallic and composite structures

Lecture 18. Airframe Maintenance Programme

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18.1 Accessibility and inspection methods

INTRODUCTION

FAR regulations

FAR 25.571 Amdt. 132

Reminder

(a) *General.* An evaluation of the strength, detail design, and fabrication must show that **catastrophic failure due to fatigue, corrosion, manufacturing defects, or accidental damage**, will be avoided throughout the operational life of the airplane. This evaluation must be conducted in accordance with the provisions of paragraphs (b) and (e) of this section,..., for each part of the structure that could contribute to a catastrophic failure...

(3) Based on the evaluations required by this section, **inspections or other procedures must be established**, as necessary, **to prevent catastrophic failure**,...

(b) **Damage-tolerance evaluation.** The evaluation must include **a determination of the probable locations and modes of damage due to fatigue, corrosion, or accidental damage**. Repeated load and static analyses supported by test evidence and (if available) service experience must also be incorporated in the evaluation.

INTRODUCTION

From the first design phases, it is important to consider that the structure must be inspected periodically to reveal possible damages (due to fatigue, corrosion, or accidental damages).

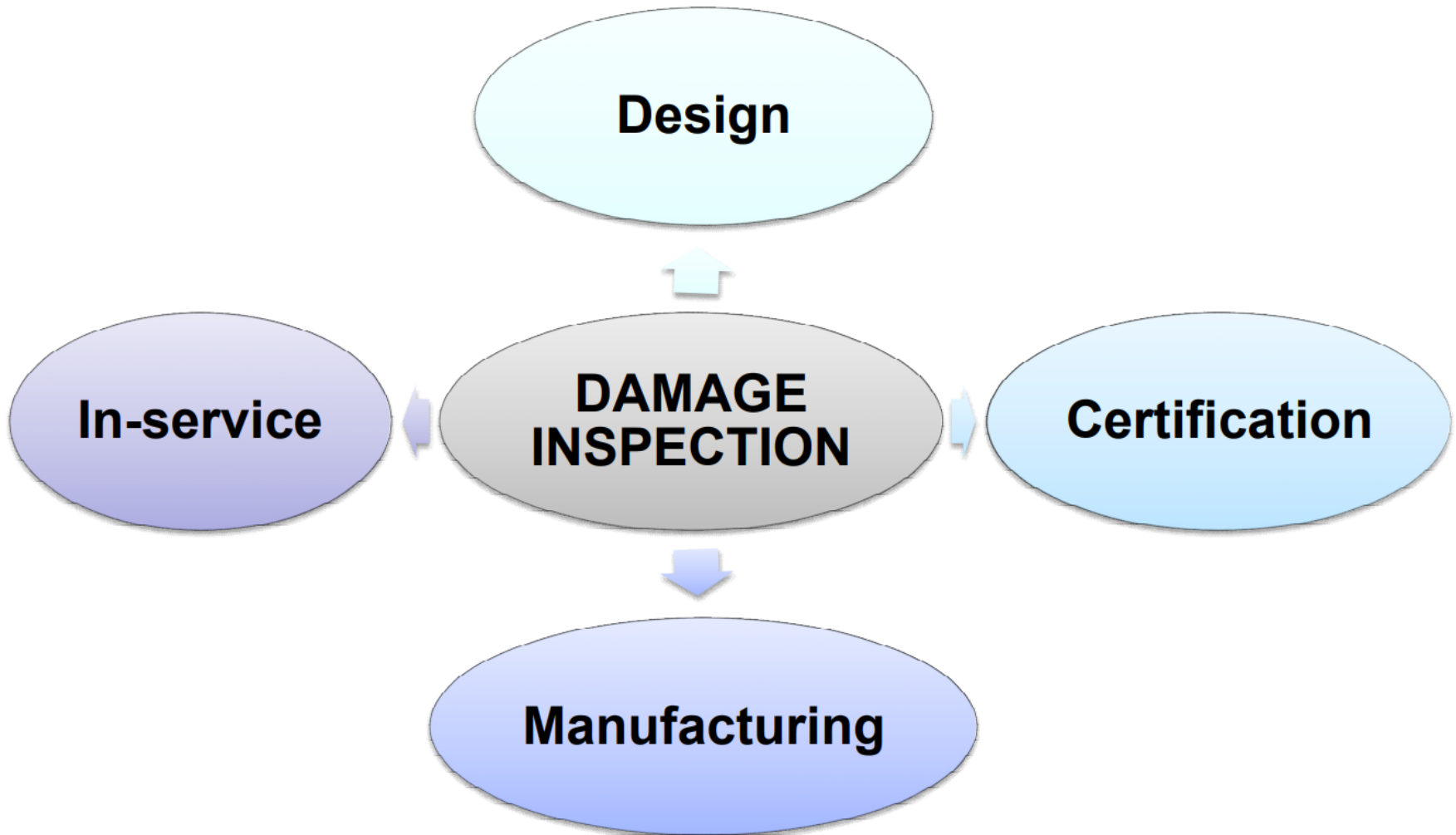
All the structural elements must be accessible somehow. This makes necessary to design specific accesses to consider the maintenance requirements that are defined. Besides, if we take into account that the corrosion and the fatigue are in general superficial phenomena, in the case of panels, it is not only necessary to access them, but to do it from both faces.

INTRODUCTION

Accesses must be anticipated not only for structural inspections, but also for system maintenance or ease of assembly. Typical examples are the following:

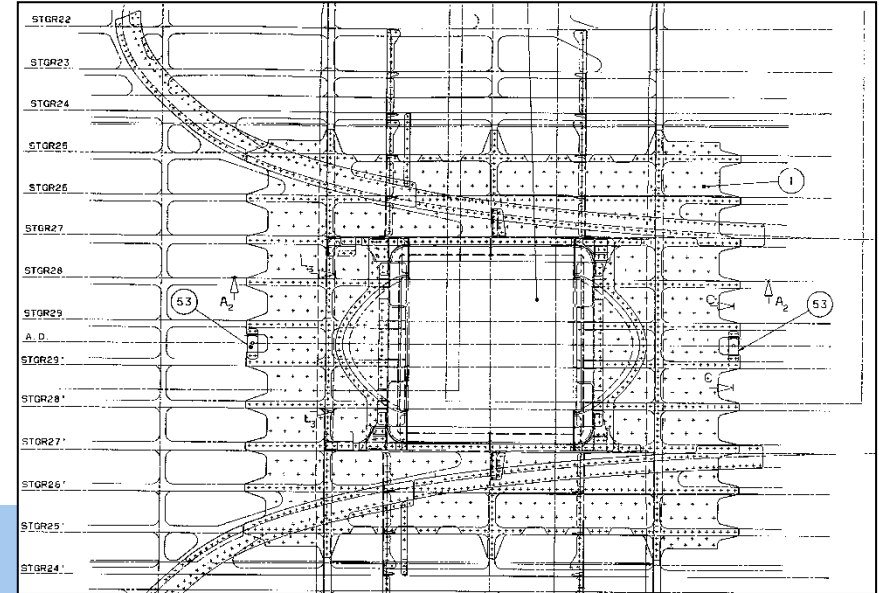
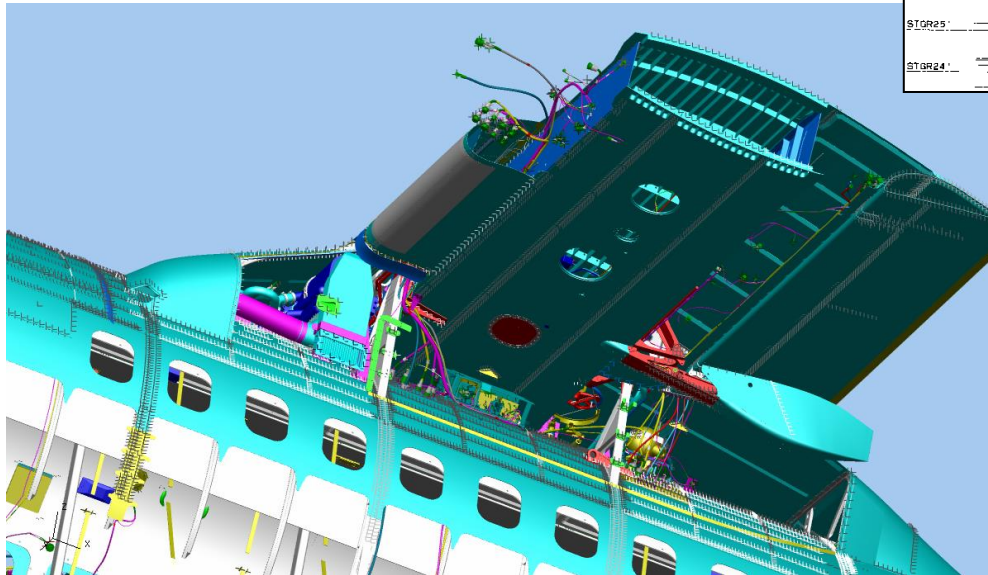
- Fuselage:
 - Access panel to the avionics compartment.
 - Access to the tail cone.
- Wing:
 - Manholes.
 - Leading edge can be disassembled.
 - Folded panels on the trailing edge.
- Stabilizers:
 - Leading edge can be disassembled.
 - Folded panels on the trailing edge.
 - Inspection holes on spars.

APPLICATION IN AIRFRAME STRUCTURES

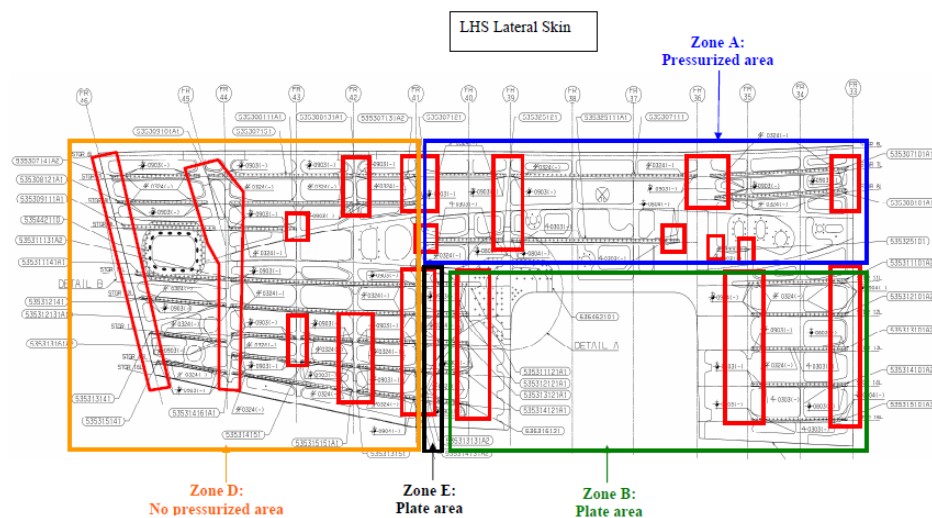


DESIGN

- Accessibility
- Ease of inspection
- Reliability
- Cost



CERTIFICATION

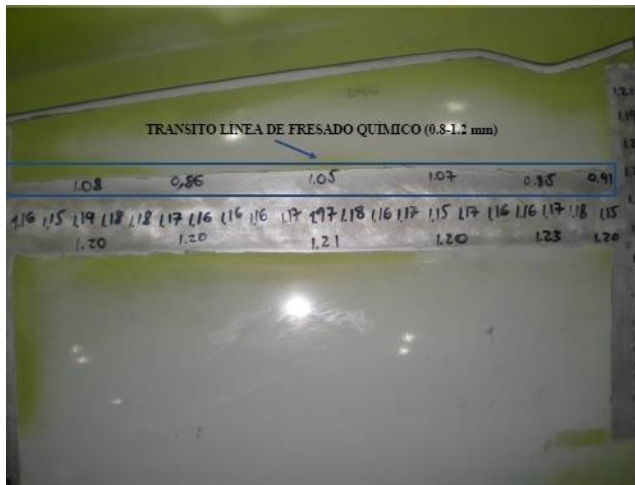


6.8.10.2 Inspection Frequencies, Methods and Areas

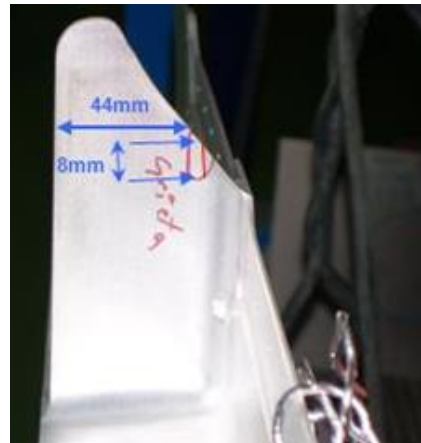
	5 000	10 000	15 000	20 000	25 000	30 000	35 000	40 000 + 10 000	
ZONE A (Current PA)				Detailed Visual	Detailed Visual	Detailed Visual	Detailed Visual	Detailed Visual	HFEC
ZONE B (plate area)				HFEC	Detailed Visual	HFEC	Detailed Visual	HFEC	Ultrasonic
ZONE C					Detailed Visual	Detailed Visual	Detailed Visual	Detailed Visual	Ultrasonic
ZONE D (NPA)				Detailed Visual		Detailed Visual			Detailed Visual
ZONE E				HFEC	Detailed Visual	HFEC	Detailed Visual	HFEC	HFEC
ZONE F				Detailed Visual	Detailed Visual	Detailed Visual	Detailed Visual	Detailed Visual	Ultrasonic

MANUFACTURING

- Quality control (materials and elementals)
- Control and improvement of processes (Machining, welding, ...)
- Control of thermal treatments and surface treatments
- Verification of assemblies.
- Verification of repairs. (Thickness measurement)



Skin thickness measurement. US.



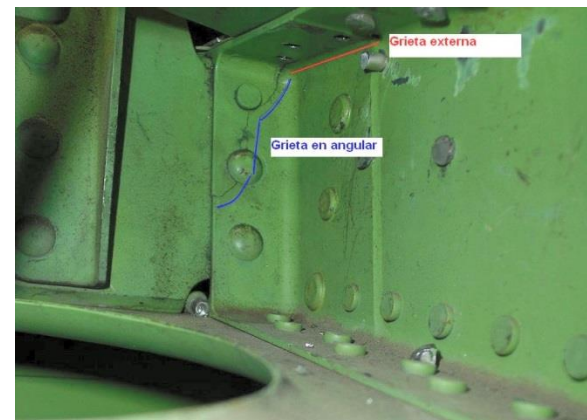
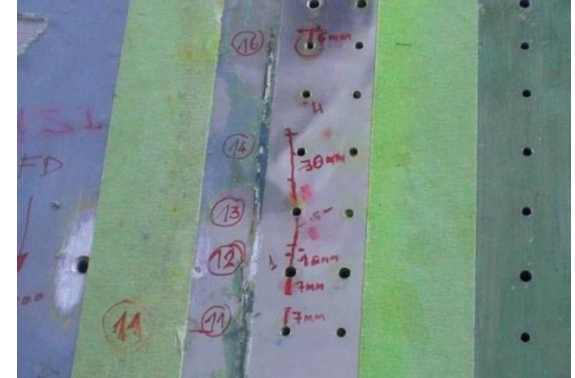
Crack in whindshield frame flange. LPI.



Engine mount welding.

IN-SERVICE

- Cracks
- Corrosion
- Erosions/Wears
- Heat damage
- Lightning damage
- Delamination



TYPES OF INSPECTIONS

General Visual Inspection (GVI)

- visual examination to detect obvious damage, failure or irregularity
- within touching distance
- under normally available lighting conditions

Detailed Visual Inspection (DET)

- intensive visual examination of a specific structural area
- available lighting supplemented with a direct source of good lighting
- inspection aids such as mirrors, magnifying lenses, etc.
- surface cleaning and elaborate access procedures may be required

Special Detailed Inspection (SDI)

- intensive examination of a specific structural area
- use of specialised inspection techniques and/or equipment
- Intricate cleaning and substantial access or disassembly necessary

SPECIAL DETAILED INSPECTIONS (SDI). NON DESTRUCTIVE TESTS (NDT)

The one that for its application uses specific equipment, instruments or materials, qualified personnel and specific certification and that, without damaging the element or structure, allows the detection and evaluation of discontinuities or defects.

Eddy currents

Ultrasonic

Liquid Penetrant

Magnetic Particles

X-Ray

Tap Testing

Visual (Boroscopic)

Thermography

Laser Interferometry

Flux Leakage

Magnetic Measurements

Acoustic Emission

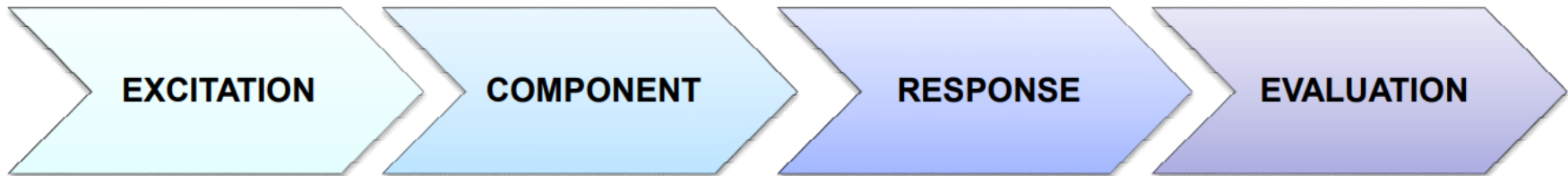
Replication

Microwave

Acoustic Microscopy

...

SPECIAL DETAILED INSPECTIONS (SDI). NON DESTRUCTIVE TESTS (NDT)



• Probe transmitter

- Mechanical vibration
- Radiation EM (EC)
- Radiation EM (RX)
- Radiation EM (Pmag)
- Radiation (Visual)

• Transmission

- Material
- Defects

• Signal detection

- Piezoelectric (US)
- Sensor IR
- Coil (EC)
- Sensor/film (RX)

• Information about

- Defects
- Properties

NON DESTRUCTIVE TESTS (NDT) IN AIRFRAME STRUCTURES**METALS**

- **Eddy currents**
- **Ultrasounds**
- Liquid penetrant inspection
- X-Ray
- Magnetic particles

COMPOSITES

- **Ultrasounds**
- **Tap Coin**
- X-Ray

(see ANNEX for more detail about NDT methods)

Flaw Type	Inspection Method	Visual	Liquid Penetrant	Magnetic Particle (A)	Ultrasonic		Eddy Current (B)	X-Ray
					Straight Beam	Angle Beam		
Surface Breaking Linear		1	3	3	1	2	3	1
Surface Breaking Volumetric Defect		3	3	3	3	3	3	3
Near-Surface Linear & Normal to Surface		0	0	2	1	2	3	1
Near-Surface, Linear & Parallel to Surface		0	0	0	3	3	0	0
Near-Surface, Volumetric		0	0	2	3	3	3	3
Subsurface, Linear & Normal to Surface		0	0	0	1	2	0	1
Subsurface, Linear & Parallel to Surface		0	0	0	3	3	0	1
Subsurface, Volumetric		0	0	0	3	3	0	3
Thickness Measurement of Thin Materials		0	0	0	3	3	3	3
Thickness Measurement of Thick Materials		0	0	0	3	3	0	3
Non-Conductive Coating Thickness Measurements		0	0	0	2	2	3	1

" Surface" refers to the surface most suitable for the inspection given the various options

(A) Ferromagnetic materials only (B) Conductive materials only

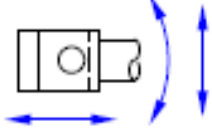
(0) Will not detect (1) Not well suited (2) Fairly well suited (3) Ideal Application

NON DESTRUCTIVE TESTING MANUAL (NDTM)

Structure of chapters similar to the SRM. (Must be used together with the SRM).

Chap. 51:

It contains general information (description NDT used in the airplane, symbology, materials, auxiliary materials, structural description of the airplane, zones).

X-RAY		RESONANCE FREQUENCY	
ULTRASONIC		MAGNETIC PARTICLE	
EDDY CURRENT		VISUAL	
PENETRANT		THERMOGRAPHY	
PROBE SCAN MOVEMENT		SEARCH UNIT SCAN / SWIVEL MOVEMENT	

NON DESTRUCTIVE TESTING MANUAL (NDTM)

Chap. 51 (cont.):

It includes other types of procedures that require application of NDT:

51-10-00 COATING THICKNESS MEASUREMENT – GENERAL

51-10-02 INSPECTION TO VERIFY THE REMOVAL OF CORROSION

51-10-10 X-RAY INSPECTION TO DETECT WATER IN HONEYCOMB SANDWICH PARTS

51-10-11 INSPECTION FOR CRACKS IN MULTI-LAYERED JOINT AL-ALLOY STRUCTURES

51-10-12 GENERAL PROCEDURE FOR THE DETECTION OF HEAT DAMAGE USING EDDY CURRENT CONDUCTIVITY MEASUREMENT

51-10-18 ROTATING PROBE TESTING OF FLAT BOTTOM HOLES IN ALUMINUM ALLOYS MATERIALS

51-10-25 THERMOGRAPHY INSPECTION TO DETECT TRAPPED WATER IN HONEYCOMB STRUCTURES MOUNTED VERTICALLY

51-10-27 VISUAL ENDOSCOPE INSPECTION FOR INTERNAL HONEYCOMB CORE SANDWICH STRUCTURES

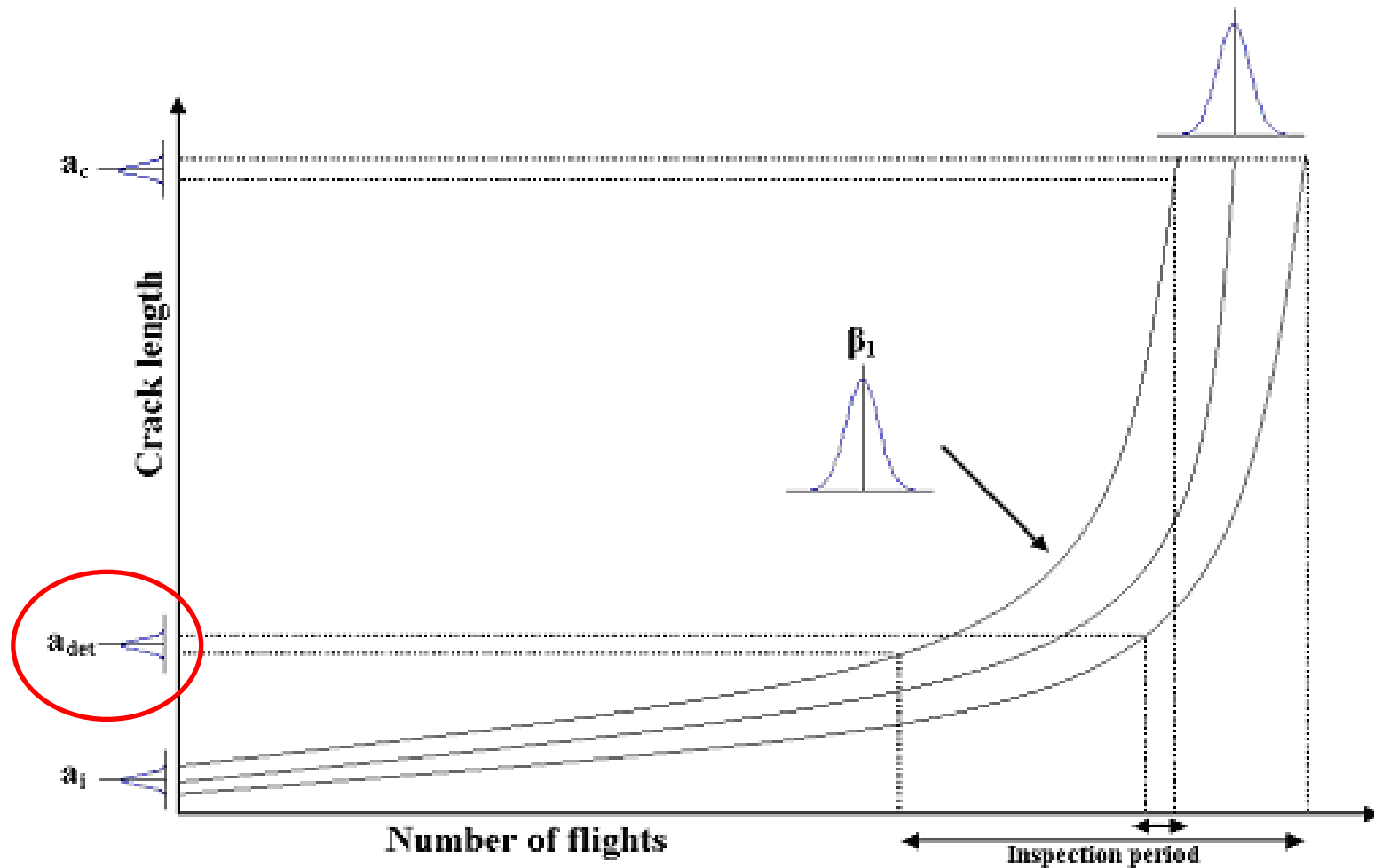
...

Chap. 52 to 57: It includes specific procedures with detailed instructions to perform the tasks of structures of the MRBD that are SDET or DET. The procedures are developed by NDT specialists with Fatigue support.

SENSITIVITY AND PROBABILITY OF DETECTION. THE WRONG MYTHS

- After applying an NDT with negative result there are no cracks for a specific application.
- The capability of an NDT is defined by the smallest detectable crack.
- All defects of equal size are of the same nature.
- All cracks are equals, whatever is their origin.
- A crack will always be detectable anywhere (laboratory / factory / field).
- A crack will always be detectable in any zone / element / location.
- All inspectors have the same capabilities / abilities

SENSITIVITY AND PROBABILITY OF DETECTION



SENSITIVITY AND PROBABILITY OF DETECTION. POD CURVE

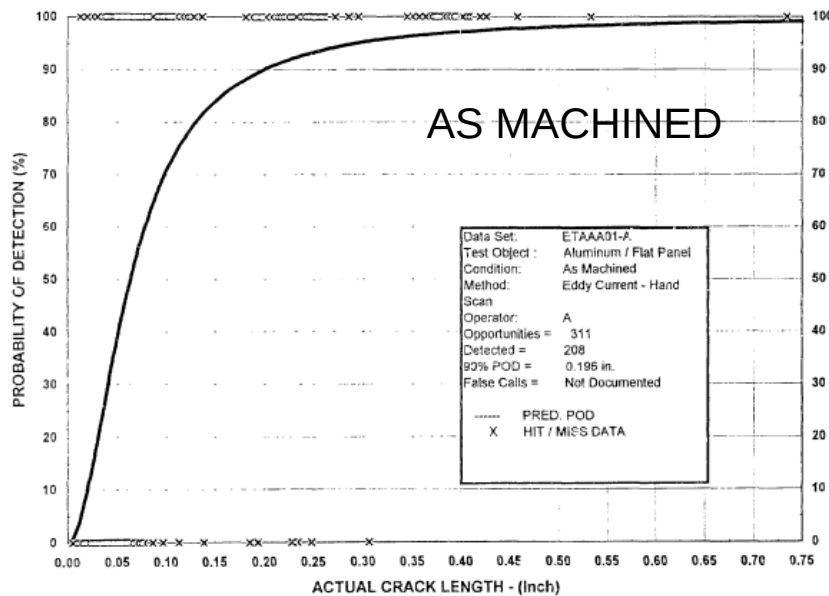
- It is the method used to measure the capability of a NDT.
- One POD and POFC per method / specimen / test conditions.
- The NDT is made on a large number of cracks of different sizes (and on areas without defects) and it is noted if something is detected or not with a certain threshold. "Hit / Miss / False Alarms"
- The results are adjusted to a model / statistical curve representative of the type of test analyzed.
- The **lower limit of 95% confidence** is plotted on that curve.
- The detectable size that characterizes the method is usually that which reaches a 90% probability with 95% confidence

SENSITIVITY AND PROBABILITY OF DETECTION. POD CURVE

Specimen: Flat plate (3.5 in x 16 in) Al2219T87 (0.06 and 0.225 in thickness).

Defects: 311 fatigue cracks at both sides.

NDT: Eddy Currents. Contact probe 100 kHz.



Inspector A: $a_{90\%} = 0.196$ in (4.97 mm)

Inspector B: $a_{90\%} = 0.184$ in (4.67 mm)

Inspector C: $a_{90\%} = 0.295$ in (7.49 mm)

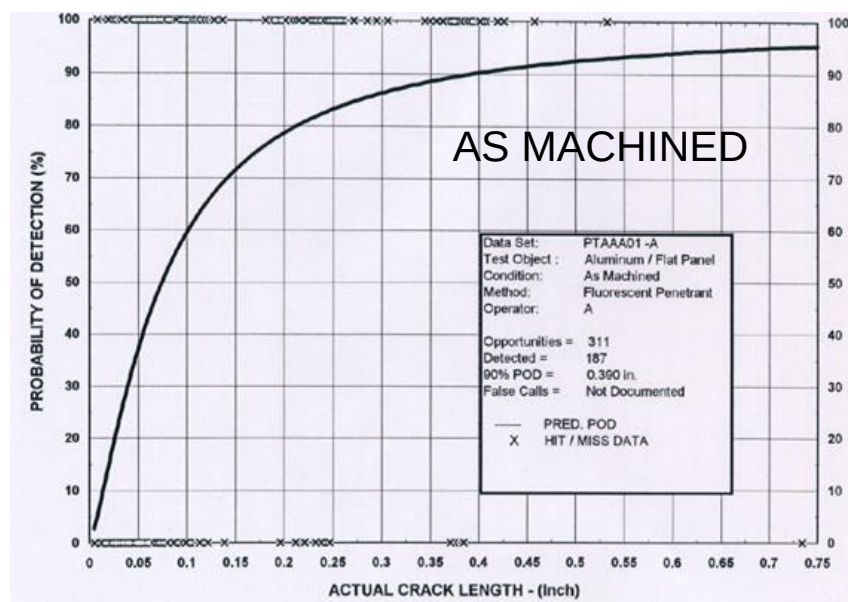
Ref. Rummel, W.D. and Matzkanin, G. A., Nondestructive Evaluation (NDE) Capabilities Data Book, NTIAC #DB-95-02, May 1996.

SENSITIVITY AND PROBABILITY OF DETECTION. POD CURVE

Specimen: Flat plate (3.5 in x 16 in) Al2219T87 (0.06 and 0.225 in thickness).

Defects: 311 fatigue cracks at both sides.

NDT: Fluorescent Penetrant Manual, Solvent Removable, Spray Developer.



Inspector A: $a_{90\%} = 0.39$ in (10 mm)

Inspector B: $a_{90\%} = 0.435$ in (11 mm)

Inspector C: $a_{90\%} = 0.313$ in (8 mm)

AFTER ETCH

Inspector A: $a_{90\%} = 0.089$ in (2.25 mm)

Inspector B: $a_{90\%} = 0.077$ in (1.96 mm)

Inspector C: $a_{90\%} = 0.106$ in (2.7 mm)

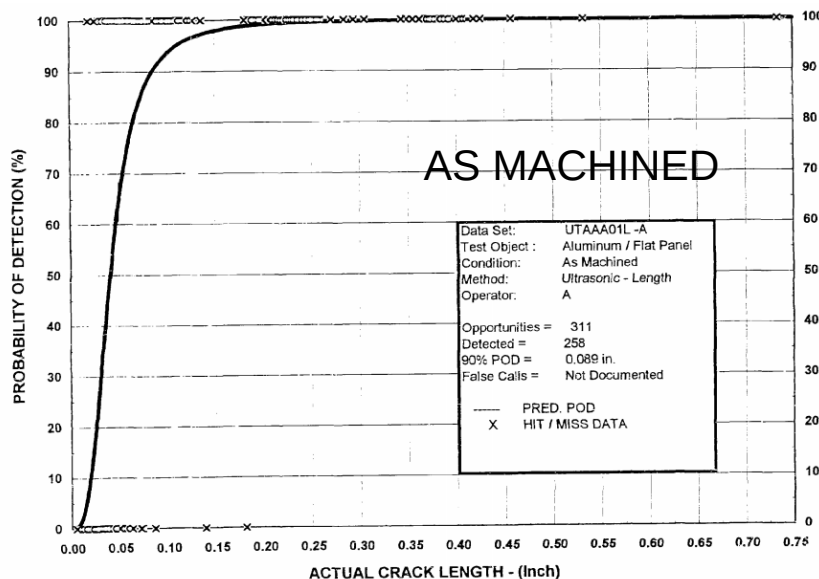
Ref. Rummel, W.D. and Matzkanin, G. A., Nondestructive Evaluation (NDE) Capabilities Data Book, NTIAC #DB-95-02, May 1996.

SENSITIVITY AND PROBABILITY OF DETECTION. POD CURVE

Specimen: Flat plate (3.5 in x 16 in) Al2219T87 (0.06 and 0.225 in thickness).

Defects: 311 fatigue cracks at both sides.

NDT: Ultrasonic surface wave.



Inspector A: $a_{90\%} = 0.089$ in (2.25 mm)

Inspector B: $a_{90\%} = 0.092$ in (2.33 mm)

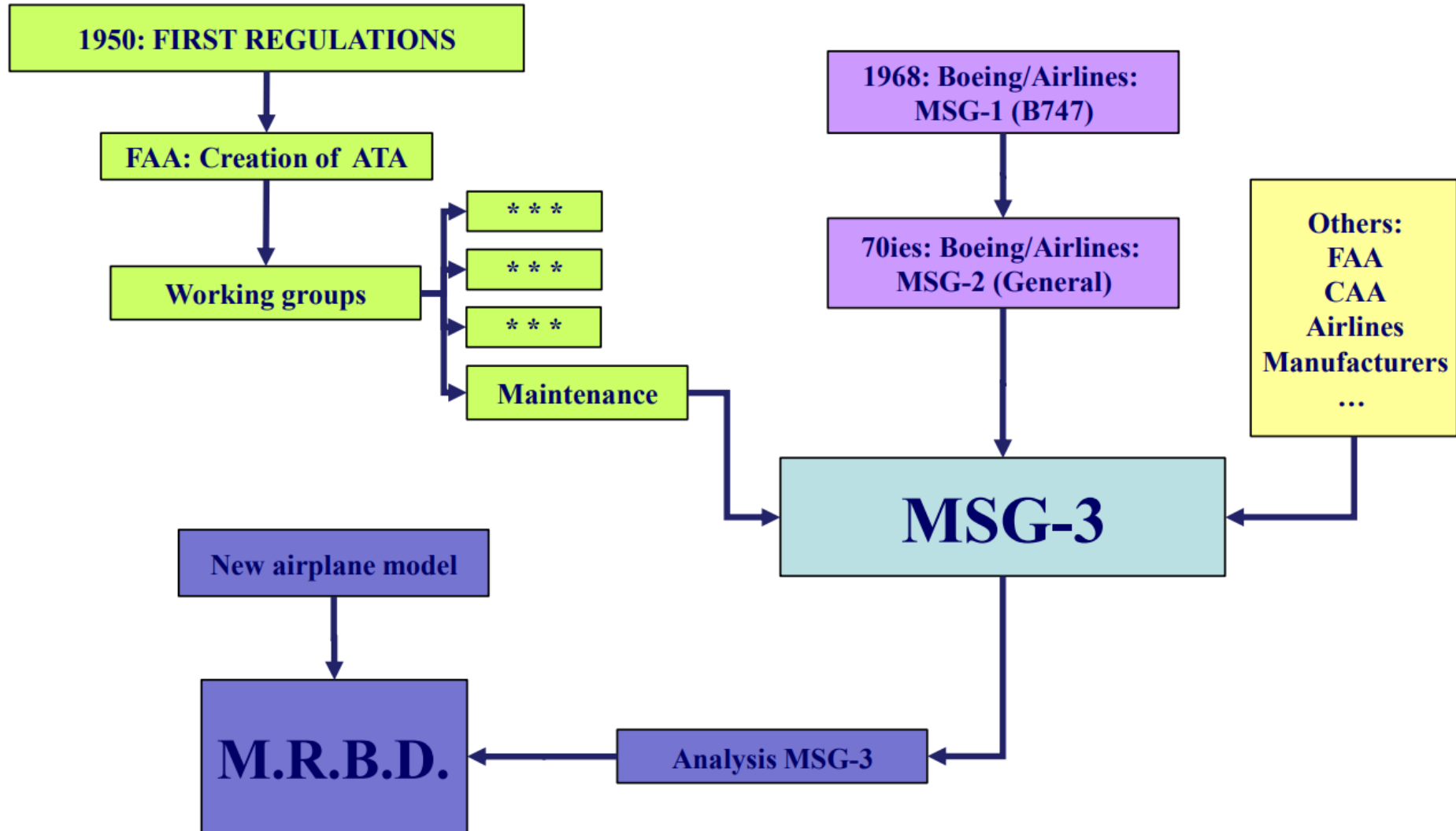
Inspector C: $a_{90\%} = 0.088$ in (2.23 mm)

Ref. Rummel, W.D. and Matzkanin, G. A., Nondestructive Evaluation (NDE) Capabilities Data Book, NTIAC #DB-95-02, May 1996.

SENSITIVITY AND PROBABILITY OF DETECTION. CONCLUSIONS.

- Importance of design (accessibility / ease of inspection / cost of inspection)
- Detectabilities are not 100% probable values and depend on many factors for the same method. It is best to ask the experts.
- There is always a better method than others to apply in a certain inspection. Again, it is better to consult the experts.
- Thickness can be measured with US or HFEC (paint layer) very accurately. It is necessary to define and correctly apply the procedure and make a good calibration.

18.2 Maintenance program



MSG-3

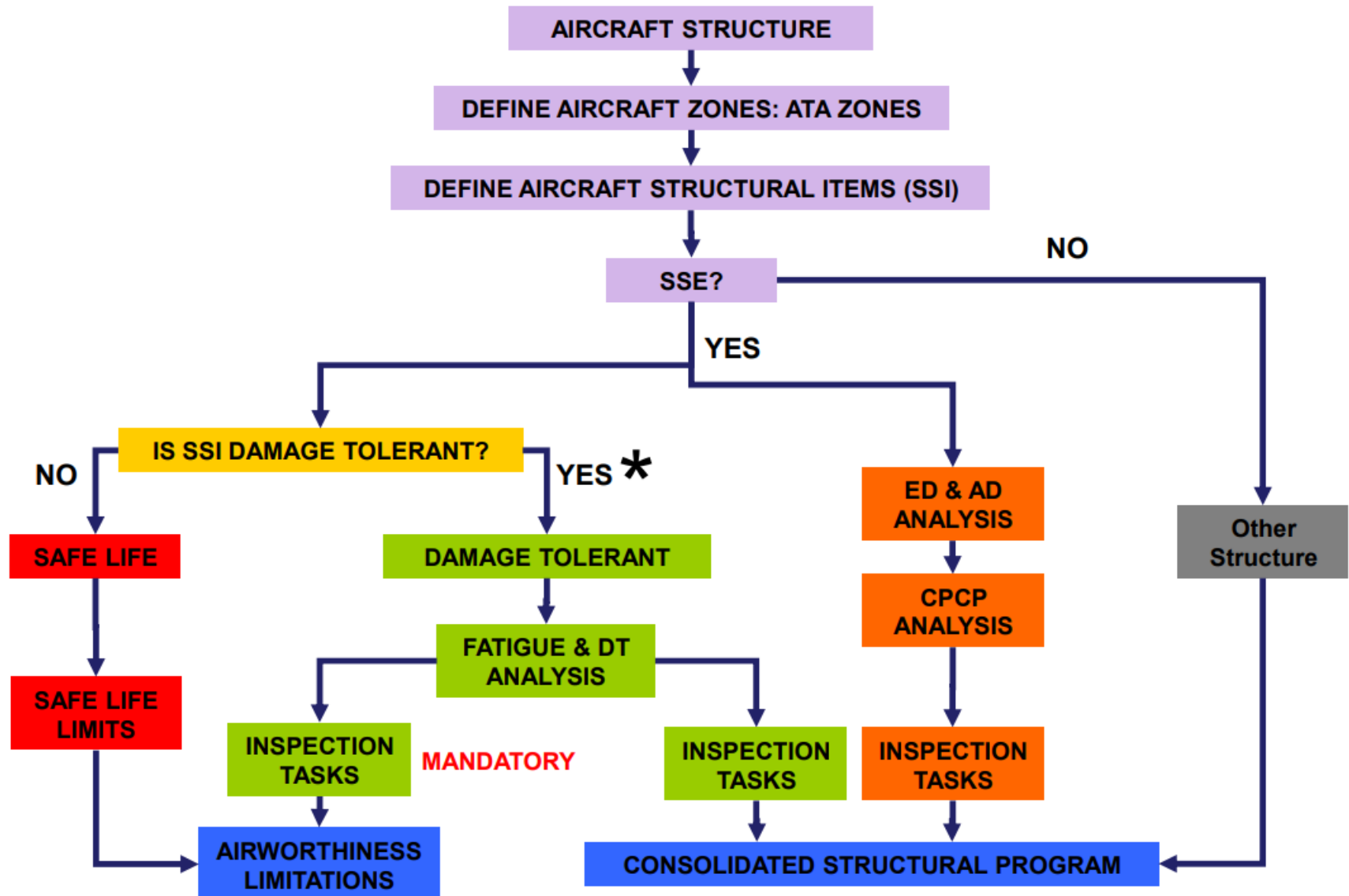
The "MSG-3" contains a logical process of analysis that for each element analyzed consists of:

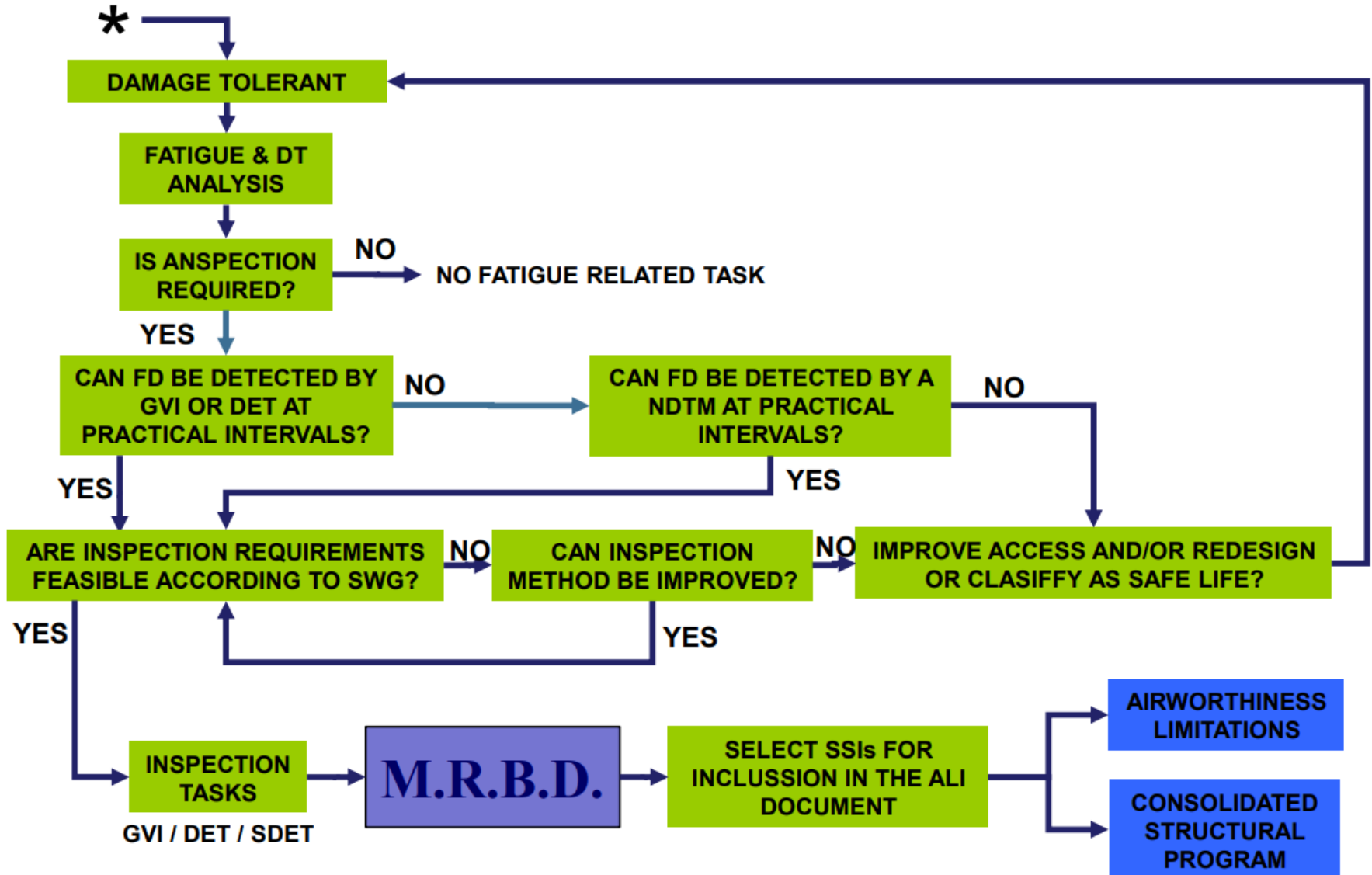
- Identify the functions performed by the element.
- Determine the possible failure modes of each of the functions.
- Determine the effects of each of these failure modes.
- Identify the possible causes of each failure.
- Determine if there is an applicable and effective scheduled maintenance task that can prevent the effects of each failure.

The "MSG-3" consists of 3 sections of analysis:

- Systems and Power Plant.
- **Structure**
- L / HIRF (Lightning and High Intensity Radiated Field)

Each of the sections indicated incorporates its own explanatory material and the corresponding logical decision diagrams. Each of the sections can be used independently of the others.





MAINTENANCE PROGRAM PREPARATION

The goal of the maintenance program is to keep the airworthiness of the aircraft through all its service life.

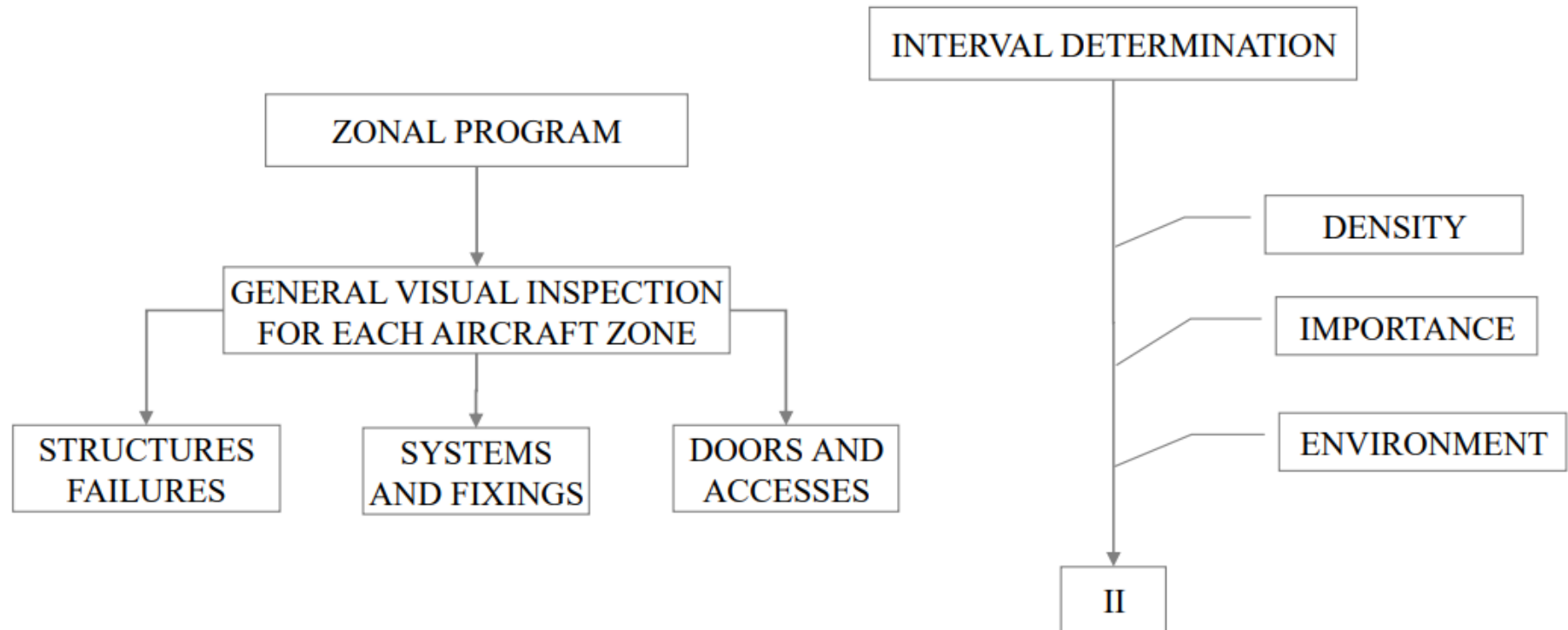
As a result of several analyses, not only of the structure, but also of the engines and systems, a series of minimum maintenance requirements are defined and written in a document called “Maintenance Review Board Document” (MRBD). This document has three parts:

Zonal Program: it includes the inspections, at a general visual level (GVI), that will be performed on the structure and the installations of all the systems.

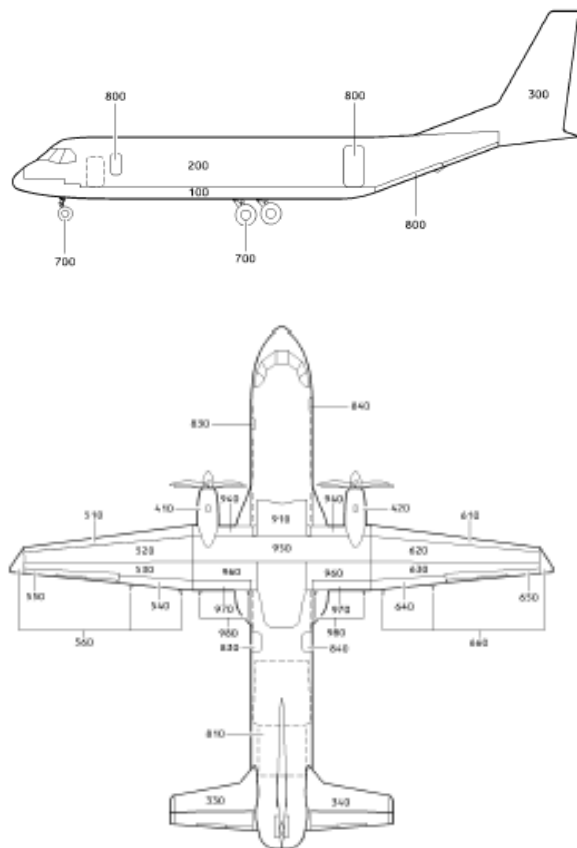
Systems and Power Plant Program: it defines the tasks to perform on each Maintenance Significant Item (MSI), considering the possible consequences of the possible failures and the causes that would produce them.

Structures Program: it includes all the inspections, at detailed or non destructive visual level (DET and SDET), required for each SSI (Structural Significant Item). These requirements are obtained from the Fatigue and damage tolerance analyses, and the environmental and accidental damages evaluation.

ZONAL PROGRAM



ZONAL PROGRAM



Task	INSPECTION					Zonal			
Number	Level	Threshold	Interval	Disc	TPS	Zone	Access	Ref.	SSI DESCRIPTION
<u>ATA 53 : FUSELAGE</u>									
53.11.04.T01	GVI	4 Years	4 Years	ALL	--	120 130	Not required	ZL100.01	Forward fuselage, external skin from Frame 1 to Frame 13 and below cabin floor level.
53.11.05.T01	GVI	4 Years	4 Years	ALL	--	210 220 230	Not required	ZL200.01	Forward fuselage, external skin from Frame 1 to Frame 13 and above cabin floor level.
53.11.14.T01	GVI	4 Years	4 Years	ALL	--	121	711, 712	ZL121.01	Forward fuselage, nose landing gear well external surface.
53.21.01.T01	GVI	4 Years	4 Years	ALL	--	130 140 150	Not required	ZL100.01	Center fuselage, external skin from Frame 13 to Frame 30 below cabin floor level except areas under fairings.
53.21.02.T01	GVI	8 Years	8 Years	ALL	--	230 240 250	Not required	ZL200.01	Center fuselage, external skin from Frame 13 to Frame 30 above cabin floor level except areas under fairings.

STRUCTURES PROGRAM

The structure program defines the inspections required for each SSI (Structural Significant Item). These inspections are obtained from the Fatigue Analyses and the Damage Tolerance Analyses, including the environmental and accidental analyses.

The inspection levels of the structure program can be DET (Detailed Visual Inspection) or SDET (Special Detailed Inspection). The inspections that were defined in the analyses as GVI (General Visual Inspection), are transferred to the zonal program, if possible.

If, due to access or frequency reasons, any of these inspections was not covered by the zonal program, it would be transferred to DET or SDET (recalculating its frequency).

For each SSI, several inspections can be defined, depending on the damage sources to consider in each case. These inspections are defined with a frequency that can be defined in terms of flight cycles, flight hours or by the calendar time.

STRUCTURES PROGRAM

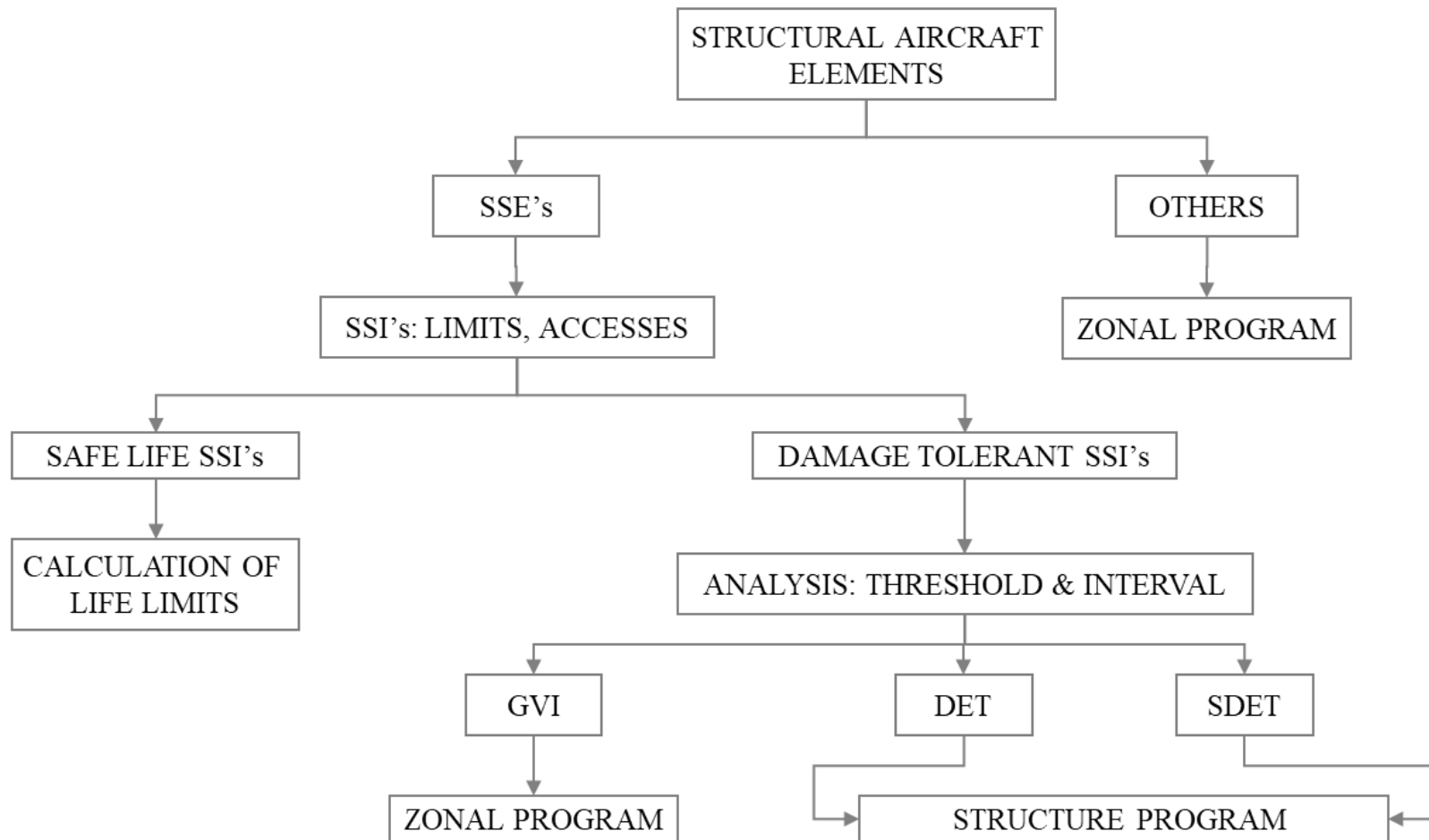
GVI inspections:

They are transferred to the zonal program.

DET and SDET inspections:

- Inspections indicated as flight cycles:
 - Fatigue (with or without accidental damage influence) of metallic materials.
- Inspections indicated by calendar time:
 - Corrosion (with or without accidental damage influence) of metallic materials.
 - Aging of composite materials.
 - Accidental damage of composite materials.

STRUCTURES PROGRAM



STRUCTURES PROGRAM (Sampling Program)

Fatigue:

Once the Fatigue inspection threshold of a SSI is calculated, it can have a very high value. Usually, the thresholds are limited to a previously selected value that is taken as the design goal. This threshold value selected is usually around 50% of the design life. When the calculated threshold is higher than the selected value, according to the analyses, no Fatigue inspections would be necessary during the period between both thresholds. Therefore, in some programs, like Airbus, a sampling program is defined so that, in the period between both thresholds, only a part of the fleet is inspected, that with the highest number of accumulated flights (Fleet Leader).

Corrosion:

For the surveillance of corrosion, when it is expected that corrosion occurs systematically (and not randomly), it is also possible to establish a sampling program that would finish when the airplane reaches its design life. In this case the airplanes with the longest service life (and not the highest number of flights) would be selected.

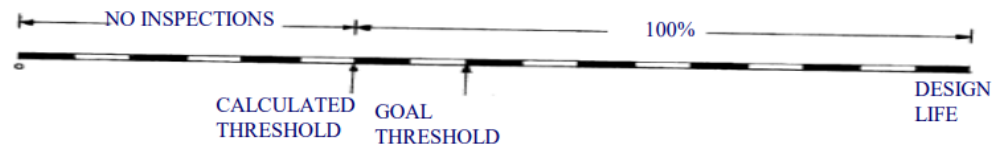
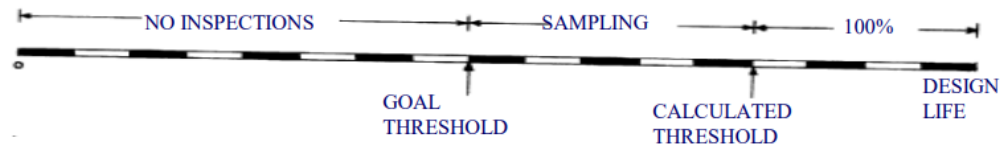
Aging:

Due to the lack of experience with composite materials, no sampling is allowed for this concept.

STRUCTURES PROGRAM (Sampling Program)

Accidental damage:

Taking into account that this type of damage can occur in any airplane, and not necessarily in those with the highest number of flights or longest service life, no sampling is allowed for those SSI's prone to accidental damage.



FATIGUE INSPECTIONS



CORROSION INSPECTIONS

AIRWORTHINESS LIMITATIONS SECTION (ALS)

The Airworthiness Limitations Section contains all the mandatory maintenance tasks of the airplane. In particular, it contains all the mandatory inspections or life limits derived from the fatigue and damage tolerance analysis of the structure.

The Airworthiness Limitations Section (ALS) belongs to the Instructions for Continuous Airworthiness (ICA) as required by FAR 25.1529. It comprises the following sections:

- **ALS Part 1:** Safe Life Airworthiness Limitation Items (SL-Ali)
- **ALS Part 2:** Damage-Tolerant Airworthiness Limitation Items (DT-Ali)
- **ALS Part 3:** Certification Maintenance Requirements (CMR)
- **ALS Part 4:** Aging System Maintenance (ASM)
- **ALS Part 5:** Fuel Airworthiness Limitations (FAL)

ALS Parts 1 and 2 have structural implications.

AIRWORTHINESS LIMITATIONS SECTION (ALS)

ALS Part 1 contains the Safe Life Airworthiness Limitations Items and provides each mandatory replacement time determined as necessary for compliance with FAR 25.571(c). Safe Life Airworthiness Limitations Items (SL-ALI) have to be considered as “service life limited components” or “life-limited parts” and therefore it is necessary to assure their replacement, traceability and monitoring in accordance with the instructions given in the ALS Part 1.

ALS Part 2 provides mandatory instructions and airworthiness limitations arising from fatigue and damage tolerance evaluation of damage tolerant structural elements whose failure could contribute to catastrophic failure of the aircraft.

DT-ALI must be inspected on all applicable aircraft with the quoted inspection level and access requirement, not later than the threshold and repeated not later than each subsequent interval.

The Threshold is the time at which the maintenance requirement is first due.

The interval for repetitive maintenance requirements is counted from the time the task was previously performed.

AIRWORTHINESS LIMITATIONS SECTION (ALS)

3.1.3 ALS PART 1 AIRWORTHINESS LIMITATIONS

Table below defines the Safe Life Airworthiness Limitations

REV. CODE	NOMENCLATURE	PART NUMBER	LIFE LIMIT (FC)	APPLICABILITY
	Nose Landing Gear	95-42005-00	20559 FC	All
	Main Landing Gear	95-41005-00	20559 FC	All

AIRWORTHINESS LIMITATIONS SECTION (ALS)

3.2.3 ALS PART 2 AIRWORTHINESS LIMITATIONS

Table below defines the Damage Tolerant Airworthiness Limitations

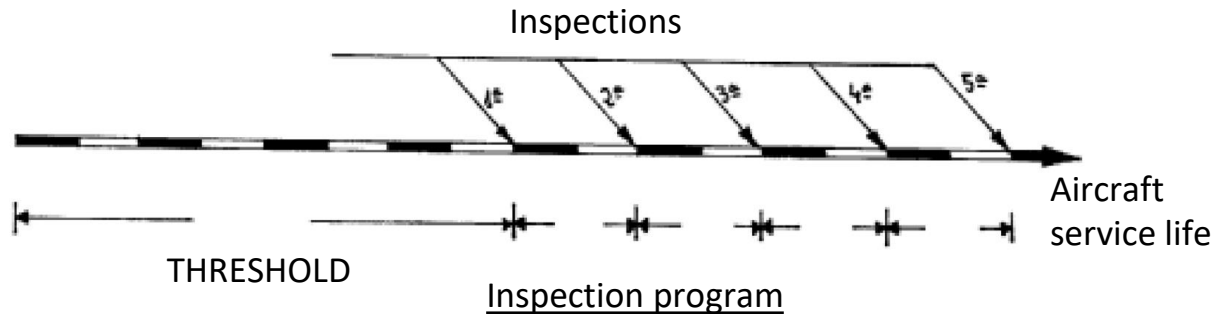
REV. CODE	SSE N°	ALI TASK REF	DESCRIPTION	THRESHOLD	INTERVAL	APPLICABILITY
	FF01		Forward Fuselage - Skins			
	FF0101	531101-01-01	Longitudinal skin splices	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL
	FF0102	531102-01-01	Circumferential skin splices	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL
	FF0103	531103-01-01	Typical basic longitudinal and transversal crack in skin	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL
	FF02		Forward Fuselage - Internal Structure Over the Floor			
	FF0201	531104-01-01	Frames	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL
	FF0202	531105-01-01	Skin riveted to stringers	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL
	FF0203	531106-01-01	Pressure floor structure	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL
	FF03		Forward Fuselage - Internal Structure Under the Floor			
	FF0301	531107-01-01	Avionic bay areas	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL
	FF04		Forward Fuselage - Pressure Bulkhead at FR1			
	FF0401	531108-01-01	Frame 1	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL
	FF0402	531109-01-01	Riveted joint FR1 to fuselage skin	15000 FC OR 21888 FH	5560 FC OR 8110 FH	ALL

18.3 Inspection frequencies

INTRODUCTION

Inspection threshold: defines when the first inspection should be performed

Inspection interval: defines the maximum period between two consecutive inspections



Type of damage	Threshold	Interval
Fatigue	Flights	Flights
Environmental	Years	Years
Metallic accidents	---	Flights/Years
Composite accidents	---	Years

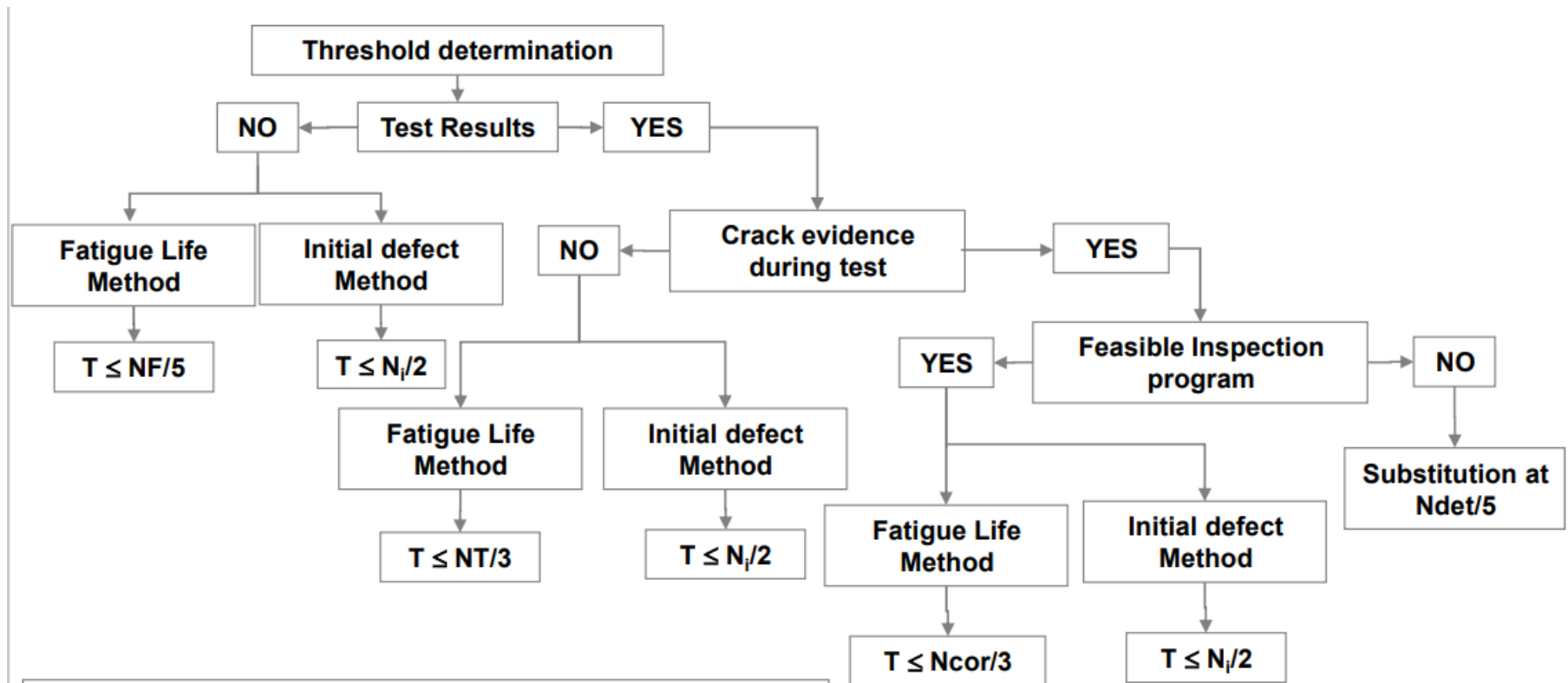
FATIGUE INSPECTIONS

To define the frequency of the Fatigue inspections (threshold and interval), the results of the analyses or tests (if present) are used.

The threshold calculation is made, for example, from the Fatigue results, either coming out from an analysis, test, or test correlated analysis. In some cases, for example in areas where the propagation of the cracks is very fast, the results of the crack growth analysis can be used.

The inspection interval is calculated from the crack growth analysis, except for those cases of multiple load path where a propagation analysis would not provide reliable results. In this latter case, the concept of residual fatigue life is used (based on the Fatigue analysis after the failure of an element).

FATIGUE INSPECTIONS. THRESHOLD DETERMINATION



NF = Fatigue life calculated by analysis

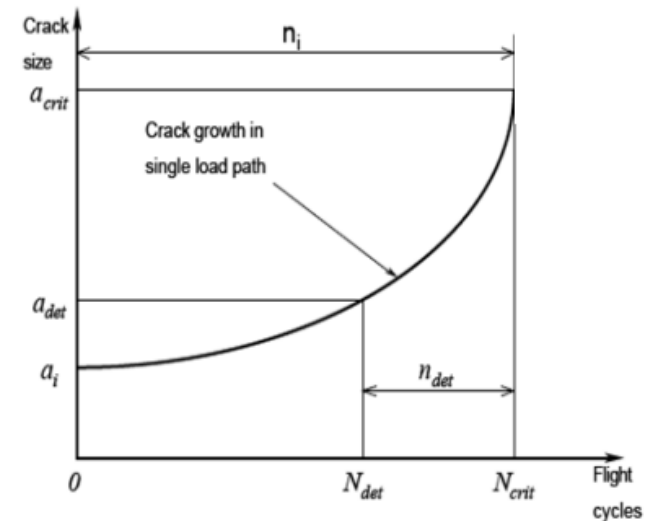
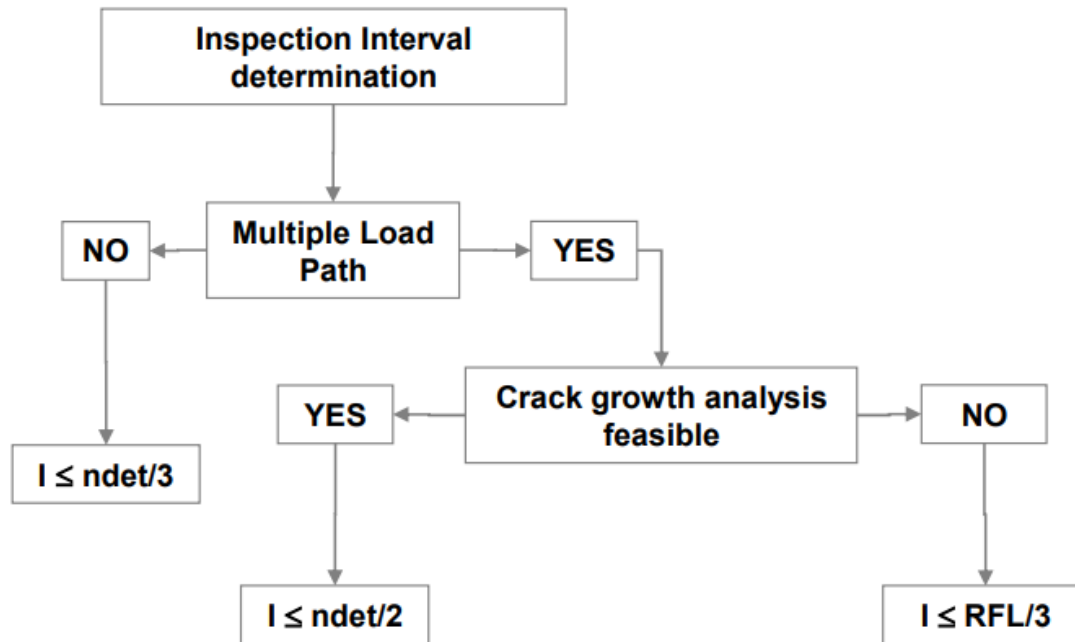
N_i = Number of flights from initial to critical crack length

NT = Fatigue life justified by test

N_{cor} = Fatigue life calculated by analysis and corrected by test

N_{det} = Number of flights in the test up to detectable crack size

FATIGUE INSPECTIONS. INSPECTION INTERVAL DETERMINATION



a_i = Length of the initial manufacturing defect.
 a_{det} = Detectable length of the crack.
 a_{crit} = Critical crack length.
 N_{det} = Number of flights to reach the detectable size.
 N_{crit} = Number of flights to reach the critical size.
 n_i = Number of flights from initial to critical size
 n_{det} = Number of flights from detectable to critical crack length
 RFL = Residual fatigue life after failure of primary load path

FATIGUE INSPECTIONS. INSPECTION INTERVAL DETERMINATION

C-295
MSG-3 LOGIC ANALYSIS
FATIGUE DAMAGE
SHEET

CASA C295	MSG-3 logic analysis	STRUCTURES	SSI No: 53-21-03
FATIGUE DAMAGE SHEET 1			

Accidental Damage Considered? Yes ☐ No ☐	Inspection Level Selected: ☐ GVI ☐ DET
---	---

Combination of Ratings Chart:

VIEWING RATING	CONGESTION RATING	LIGHTING RATING	SURFACE RATING
☐ NO ACCESS - Hidden item or distance greater than 3 meters.	Go to next higher inspect level	☐ Exterior of A/C in Shadow (e.g. landing gear bay without direct light)	Not Clean Area 1 Clean Area 2
☐ POOR - When distance is 1.5 meters to 3.0 meters.	Congested 1 Moderate 2 Clear Area 3	☐ Exterior of A/C in full day light or inside A/C with artificial light	Not Clean Area 2 Clean Area 3
☐ MODERATE - When distance is 0.5 meters to 1.5 meters.	Congested 2 Moderate 3 Clear Area 4	☐ Concentrated lighting as required	Not Clean Area 3 Clean Area 4
☐ GOOD - When unobstructed, as close as needed	Congested 3 Moderate 4 Clear Area 5		

SIZE RATING	ACCESS RATING
	1 2 3 4 5
Large Area	1 1 1 2 2 3
Medium Area/ Large Fitting	2 1 2 2 3 3
Medium Size Fitting	3 1 2 3 4 4
Small Area / Small Fitting	4 1 2 3 4 5

PRACTICALITY RATING	CONDITION RATING	1	2	3	4
1	295	205	145	100	
2	205	100	70	50	
3	145	70	35	22	
4	100	50	15	10	
5	70	22	10	8	

Basic Detectable Crack Length: L_{av} (mm)

LBAS (mm)	X	Gauge Factor	X	Edge Factor	=	LVIS (mm)	+	LHD (mm)	=	LDET (mm)

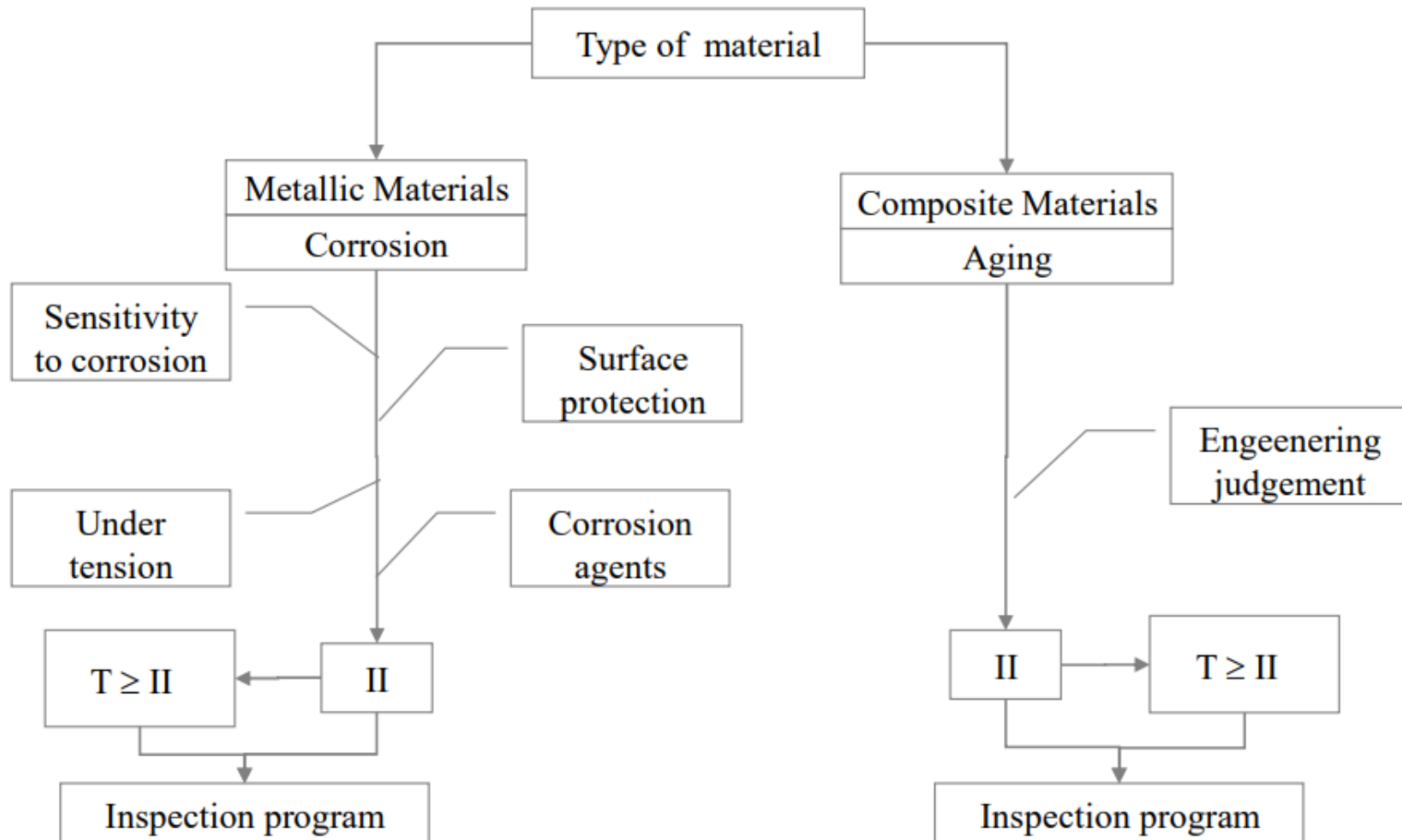
FATIGUE Inspection Task	LEVEL	THRESHOLD	INTERVAL	☐ N/A ☐ 30,000 FC ☐ 15,000 FC ☐ 10,000 FC ☐ 5,000 FC ☐ _____ FC
	☐ GVI ☐ DET			

Is this task applicable and effective?

Yes ☐ → End

No ☐ → Select Next Higher Inspection Level

ENVIRONMENTAL DAMAGE INSPECTIONS



ENVIRONMENTAL DAMAGE INSPECTIONS

C-295

MSG-3 LOGIC ANALYSIS

ENVIRONMENTAL
DETERIORATION
SHEET

	MSG-3 logic analysis	STRUCTURES	SSI No: 53-21-03
ENVIRONMENTAL DETERIORATION SHEET			

Combination of Ratings Chart:

<table border="1" style="width: 100%; border-collapse: collapse;"> <tr><th colspan="2">SELECTED MATERIAL</th></tr> <tr><td colspan="2">Clad 2024T3</td></tr> <tr><th colspan="2">TYPE OF CORROSION RATING</th></tr> <tr><td>Intergranular</td><td>1</td></tr> <tr><td>Pitting</td><td>1</td></tr> <tr><td>Uniform</td><td>2</td></tr> <tr><td>Galvanic</td><td>3</td></tr> <tr><td>Microbiological</td><td>3</td></tr> <tr><td>Filiform</td><td>3</td></tr> <tr><td>Fretting</td><td>3</td></tr> </table> Select the lowest rating number ↓	SELECTED MATERIAL		Clad 2024T3		TYPE OF CORROSION RATING		Intergranular	1	Pitting	1	Uniform	2	Galvanic	3	Microbiological	3	Filiform	3	Fretting	3	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th>ENVIRONMENTAL RATING</th> <th>PROTECTION RATING</th> <th>AD ?</th> </tr> <tr> <td rowspan="3">☐ Probable exposure to Highly corrosive products</td> <td>Average</td> <td>1</td> </tr> <tr> <td>Good</td> <td>1</td> </tr> <tr> <td>Special Attention</td> <td>2</td> </tr> <tr> <td rowspan="3">☒ Probable exposure to corrosive products</td> <td>Average</td> <td>2</td> </tr> <tr> <td>Good</td> <td>2</td> </tr> <tr> <td>Special Attention</td> <td>3</td> </tr> <tr> <td rowspan="3">☐ No Probable exposure</td> <td>Average</td> <td>2</td> </tr> <tr> <td>Good</td> <td>3</td> </tr> <tr> <td>Special Attention</td> <td>3</td> </tr> </table> ↓	ENVIRONMENTAL RATING	PROTECTION RATING	AD ?	☐ Probable exposure to Highly corrosive products	Average	1	Good	1	Special Attention	2	☒ Probable exposure to corrosive products	Average	2	Good	2	Special Attention	3	☐ No Probable exposure	Average	2	Good	3	Special Attention	3	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th>STRESS CORROSION RATING</th> <th>1</th> <th>2</th> <th>3</th> </tr> <tr> <td>HIGH</td> <td>1</td> <td>1</td> <td>2</td> </tr> <tr> <td>MEDIUM</td> <td>2</td> <td>2</td> <td>2</td> </tr> <tr> <td>☒ NOT SENSITIVE</td> <td>3</td> <td>2</td> <td>3</td> </tr> </table> → <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td>☐ 1</td> <td>☒ 2</td> <td>☐ 3</td> </tr> </table> ↓ <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td>NOTE 1</td> <td>2 Years</td> <td>4 Years</td> <td>1</td> </tr> <tr> <td>2 Years</td> <td>4 Years</td> <td>8 Years</td> <td>2</td> </tr> <tr> <td>4 Years</td> <td>8 Years</td> <td>NOTE 2</td> <td>3</td> </tr> </table>	STRESS CORROSION RATING	1	2	3	HIGH	1	1	2	MEDIUM	2	2	2	☒ NOT SENSITIVE	3	2	3	☐ 1	☒ 2	☐ 3	NOTE 1	2 Years	4 Years	1	2 Years	4 Years	8 Years	2	4 Years	8 Years	NOTE 2	3
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NOTE 1: Redesign or corrective action required.
 NOTE 2: No inspection required for Environmental Deterioration.

Resulting ED Basic Inspection Task	LEVEL THRESHOLD INTERVAL	<table style="width: 100%;"> <tr> <td>☐ GVI</td> <td>☒ DET</td> <td>☐ SDET</td> </tr> <tr> <td>☒ 8 Years</td> <td>☐ 4 Years</td> <td>☐ 2 Years</td> </tr> <tr> <td>☒ 8 Years</td> <td>☐ 4 Years</td> <td>☐ 2 Years</td> </tr> </table>	☐ GVI	☒ DET	☐ SDET	☒ 8 Years	☐ 4 Years	☐ 2 Years	☒ 8 Years	☐ 4 Years	☐ 2 Years
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CORROSION PREVENTION AND CONTROL PROGRAM REQUIREMENT ASSESSMENT

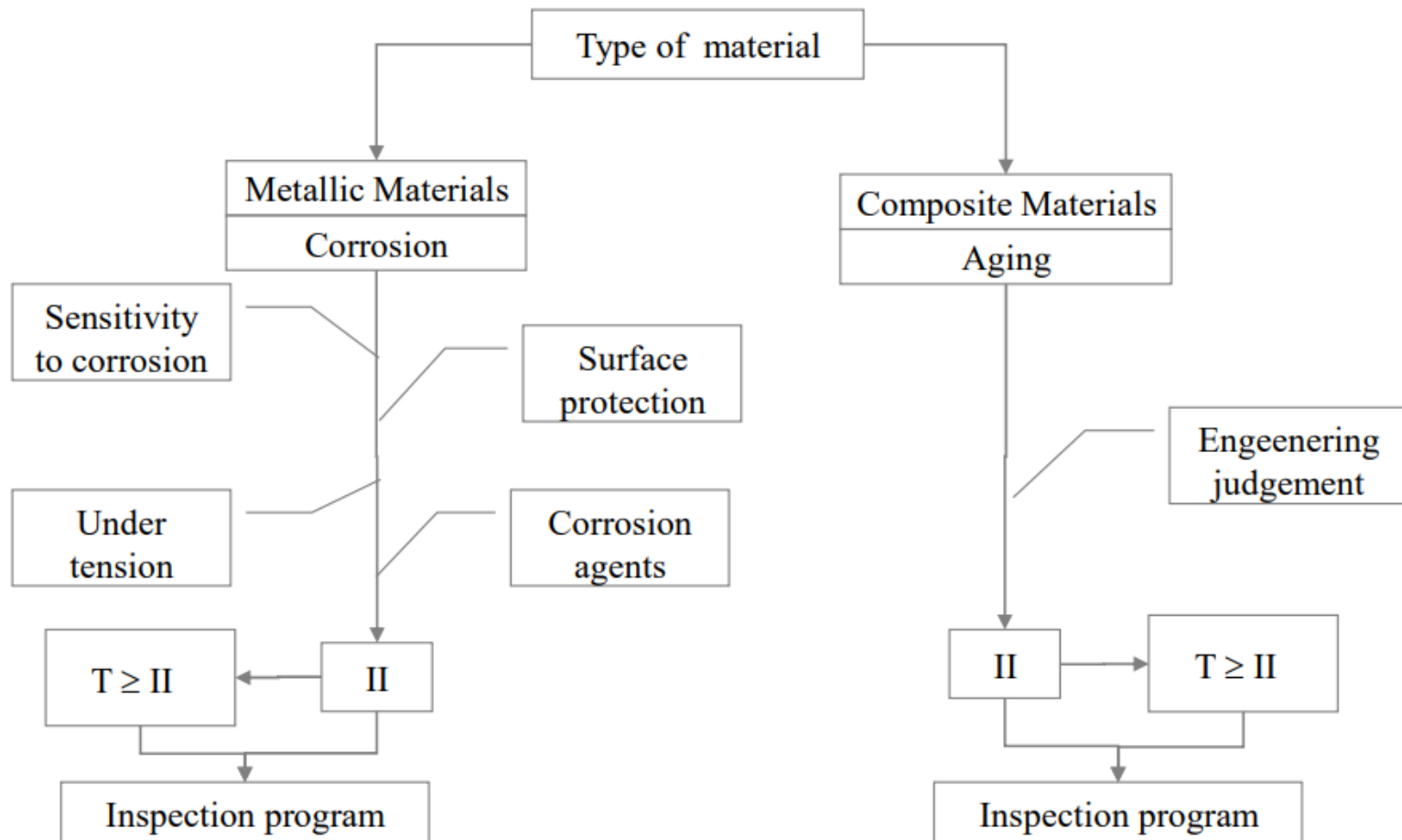
Define CPCP Basic Task:	LEVEL THRESHOLD INTERVAL	<table style="width: 100%;"> <tr> <td>☒ GVI</td> <td>☐ DET</td> <td>☐ SDET</td> </tr> <tr> <td>☐ 4 Years</td> <td colspan="2">TPS APPLICATION?</td> </tr> <tr> <td>☐ 4 Years</td> <td>☒ Yes</td> <td>☐ No</td> </tr> </table>	☒ GVI	☐ DET	☐ SDET	☐ 4 Years	TPS APPLICATION?		☐ 4 Years	☒ Yes	☐ No
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☐ 4 Years	TPS APPLICATION?										
☐ 4 Years	☒ Yes	☐ No									

Consider ED Basic Inspection Task and establish the consolidated ED/CPCP Task:

Consolidated ED/CPCP Task:	LEVEL THRESHOLD INTERVAL	<table style="width: 100%;"> <tr> <td>☐ GVI</td> <td>☒ DET</td> <td>☐ SDET</td> </tr> <tr> <td>☒ 8 Years</td> <td>☐ 4 Years</td> <td>☐ 2 Years</td> </tr> <tr> <td>☒ 8 Years</td> <td>☐ 4 Years</td> <td>☐ 2 Years</td> </tr> </table>	☐ GVI	☒ DET	☐ SDET	☒ 8 Years	☐ 4 Years	☐ 2 Years	☒ 8 Years	☐ 4 Years	☐ 2 Years
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TPS APPLICATION? ☒ Yes ☐ No

ACCIDENTAL DAMAGE INSPECTIONS



ACCIDENTAL DAMAGE INSPECTIONS

C-295

MSG-3 LOGIC ANALYSIS
**ACCIDENTAL DAMAGE
SHEET**

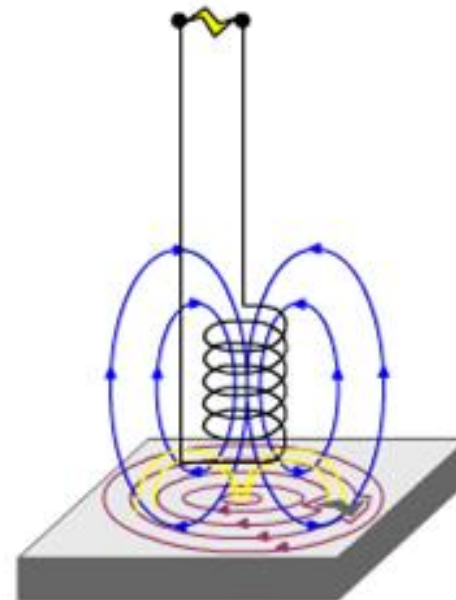
	MSG-3 logic analysis	STRUCTURES	SSI No: 53-21-03																																				
ACCIDENTAL DAMAGE SHEET																																							
Is Accidental Damage Covered by other SSI(s)? Yes ☐ → SSI(s) No: No ☒ → Continue analysis																																							
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 30%;">DAMAGE SOURCE</th> <th style="width: 5%;">LIK</th> <th style="width: 5%;">UNL</th> <th style="width: 60%;">DESCRIPTION OF DAMAGE</th> </tr> </thead> <tbody> <tr><td>Ground/Cargo Handling</td><td></td><td>X</td><td></td></tr> <tr><td>Maintenance Procedures</td><td></td><td>X</td><td></td></tr> <tr><td>Operations Procedure</td><td></td><td>X</td><td></td></tr> <tr><td>Runway Debris</td><td></td><td>X</td><td></td></tr> <tr><td>Lightning Strike</td><td></td><td>X</td><td></td></tr> <tr><td>Spillages</td><td></td><td>X</td><td></td></tr> <tr><td>Manufacturing Defects</td><td></td><td>X</td><td></td></tr> <tr><td>Others</td><td></td><td>X</td><td></td></tr> </tbody> </table>				DAMAGE SOURCE	LIK	UNL	DESCRIPTION OF DAMAGE	Ground/Cargo Handling		X		Maintenance Procedures		X		Operations Procedure		X		Runway Debris		X		Lightning Strike		X		Spillages		X		Manufacturing Defects		X		Others		X	
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METALLIC		COMPOSITE																																					
Fatigue/Corrosion Impact Assessment Is Accidental Damage to be considered in FATIGUE DAMAGE Analysis? ☐ Yes ☒ No		Select Inspection LEVEL: ☐ GVI ☐ DET ☐ SDET																																					
Is Accidental Damage to be considered in ENVIRONMENTAL DETERIORATION Analysis? ☐ Yes ☒ No		<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 20%;">SSR Rating</th> <th style="width: 80%;">Basic Interval</th> </tr> </thead> <tbody> <tr><td style="text-align: center;">0</td><td>n/a</td></tr> <tr><td style="text-align: center;">1</td><td>8 Years</td></tr> <tr><td style="text-align: center;">2</td><td>4 Years</td></tr> <tr><td style="text-align: center;">3</td><td>2 Years</td></tr> <tr><td style="text-align: center;">4</td><td>Note 1</td></tr> </tbody> </table> NOTE 1: Interval must be specified.		SSR Rating	Basic Interval	0	n/a	1	8 Years	2	4 Years	3	2 Years	4	Note 1																								
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		Is Accidental Damage likely to occur more than once during Basic Interval and Static Strength significantly reduced? No ☐ → Select Basic Interval Yes ☐ → Select Next Lower Interval Interval: 																																					

ANNEX. NDT Methods

PHYSICAL PRINCIPLE

restart

When a flaw is introduced to the conductive material, the eddy currents are disrupted.



FREQUENCY vs DEPTH OF PENETRATION

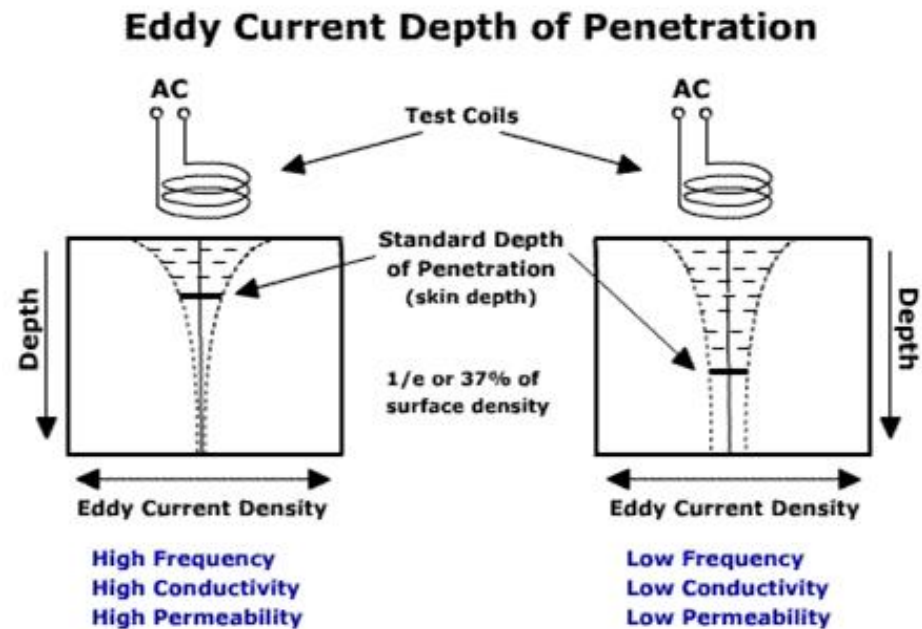
The depth of penetration depends on:

Eddy currents frequency

Material Conductivity

Magnetic permeability of the material

Standard penetration depth: the one at which the current density has dropped by a value equal to $1/e$ ($\sim 37\%$)



HIGH FREQUENCY EDDY CURRENTS (HFEC & Rototest):

$F > 10 \text{ KHz}$

Surface discontinuities

Detectabilities:

- 2-3 mm (HFEC)
- 1 mm (Rototest)

LOW FREQUENCY EDDY CURRENTS(LFEC):

$F < 10 \text{ KHz}$

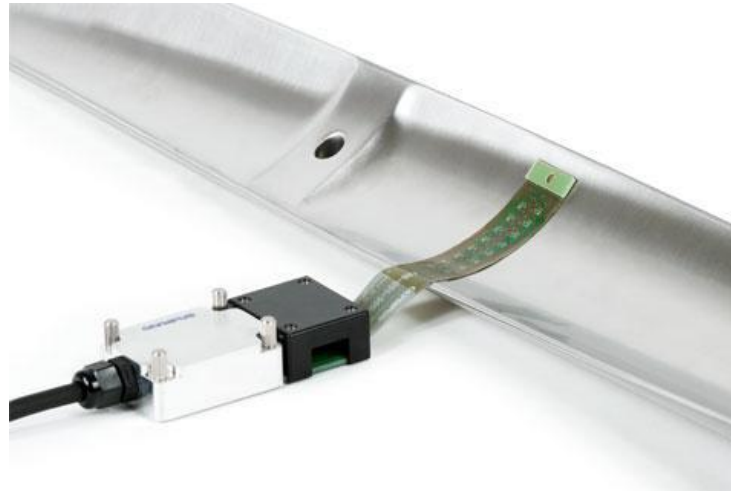
Surface discontinuities or surface defects in parts hidden by other elements.

Detectabilities between 5-10 mm

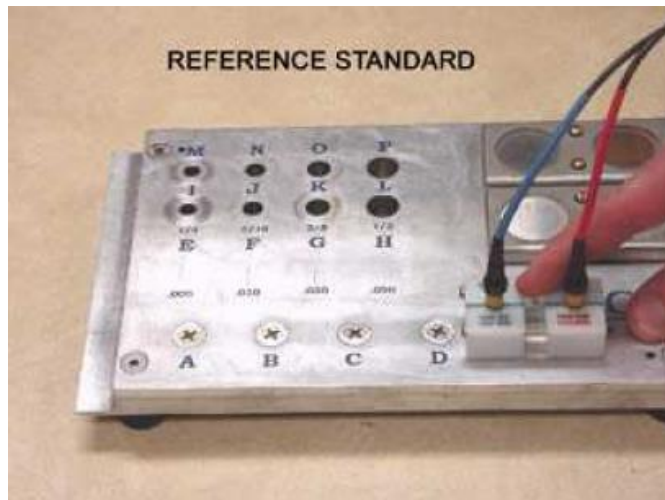
MAIN APPLICATIONS



MAIN APPLICATIONS

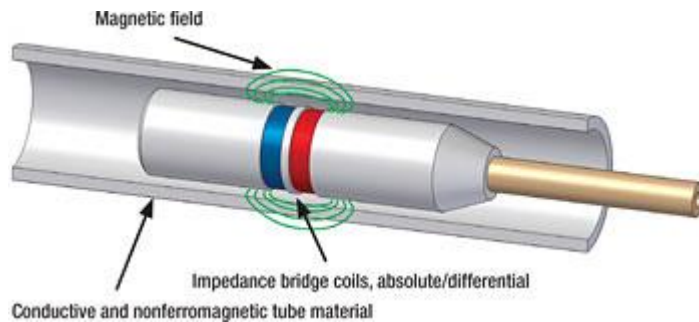


Flexible probe
Special designs



Sliding probe

MAIN APPLICATIONS



ID tube inspection probe



OD tube inspection probe.



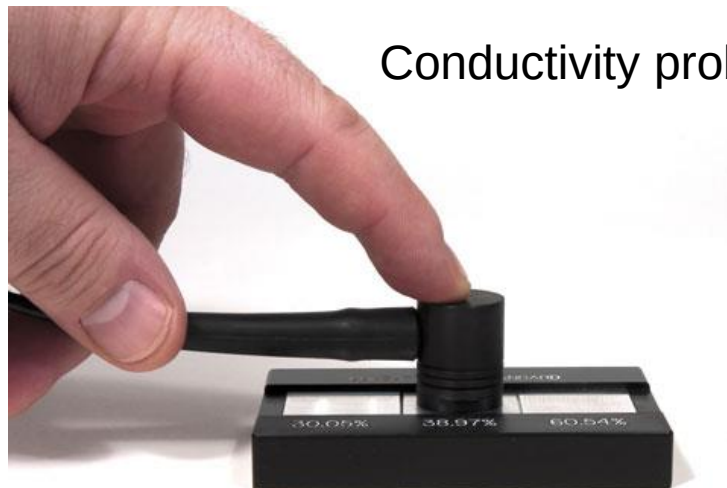
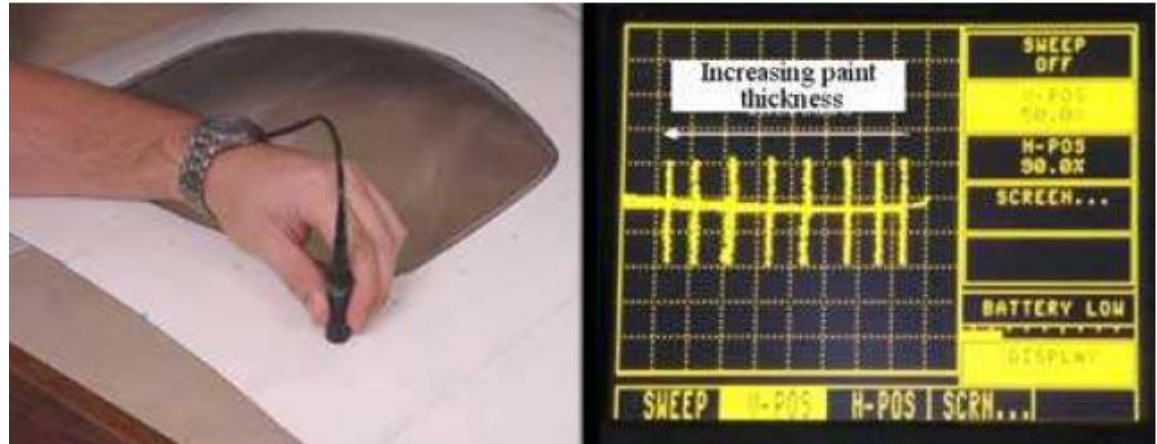
Special designs



Eddy Current Array probe.
Subsurface corrosion

MAIN APPLICATIONS

Non conductive coating
thickness measurements



Conductivity probe

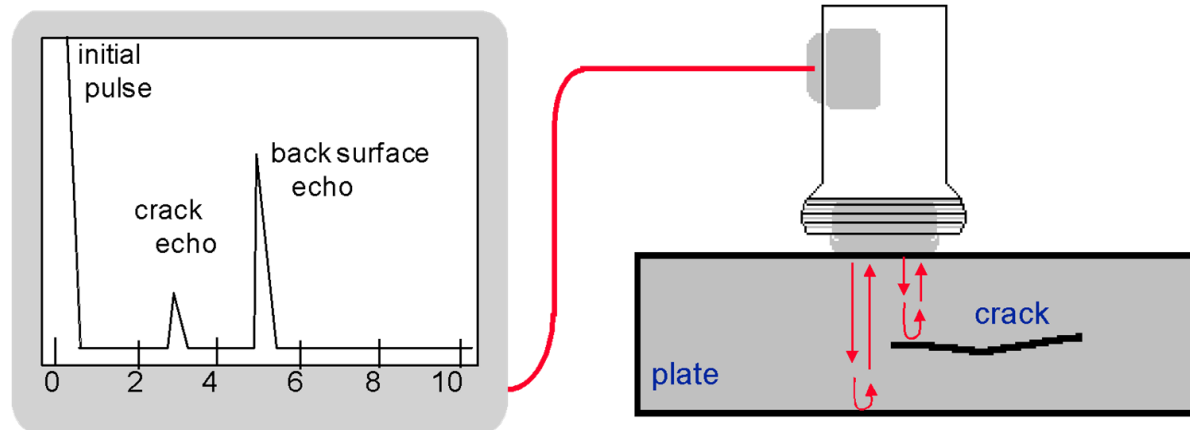


Detector

ADVANTAGES	DISADVANTAGES
— Sensitive to small cracks. — Surface defects close to the surface (HFEC). — Subsurface defects or under covered pieces (LFEC) — It provides immediate results — Equipment very easy to handle — Minimum preparation of the element to inspect — parts with complex shapes can be inspected.	— Only conductive materials — The depth of penetration is limited. — The discontinuities parallel to the inspection surface are not detected. — Surface roughness interferes with sensitivity. — Piece patterns are necessary.

PHYSICAL PRINCIPLE

- It is based on the property that sound waves have to reflect.
 - Introduction of high frequency elastic waves in a material, these waves are propagated through the material.
 - When the sonic waves reach the edge of the material it is reflected, emitting an echo.
 - Defects or cracks alter the propagation of waves.
 - The techniques that can be used are pulse echo, resonance and transmission.
- The most used echo pulse



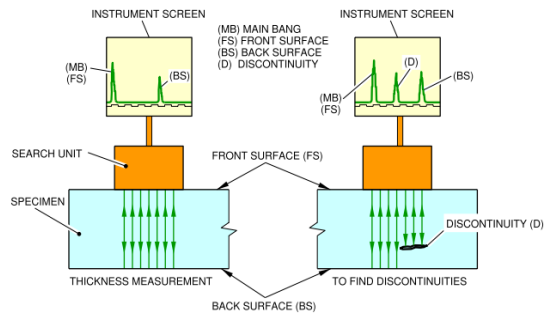
EQUIPMENT

- Equipment consists of: Electronic excitation system, transmitter/receiver probe, amplifier ...
- Coupler: water (local or complete immersion), oil, gels ..

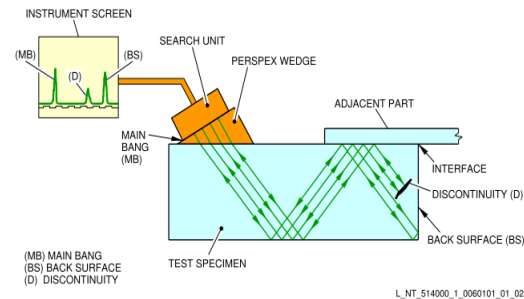


TYPES OF WAVES

Longitudinals



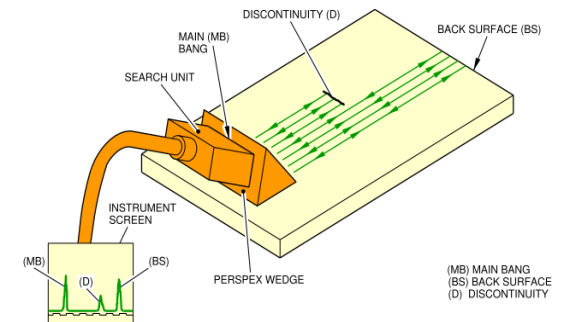
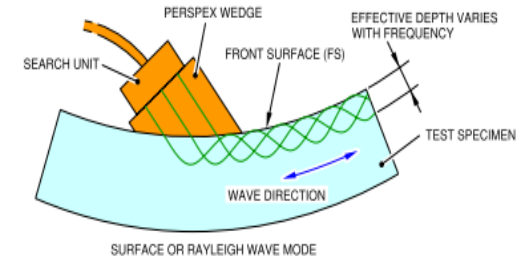
Transversals



L_NT_514000_1_0060101_01_02



Superficials



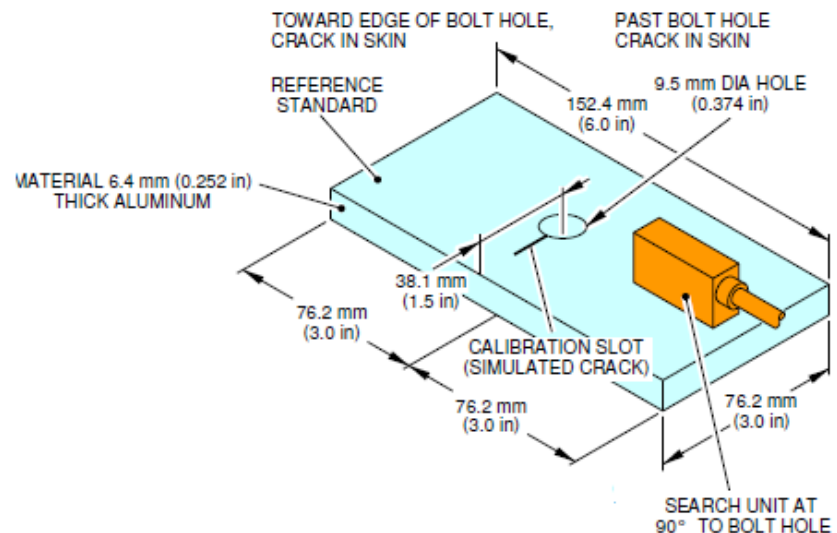
REFERENCE PATTERNS

Purpose

- Verify the inspection technique
- Adjust and calibrate the parameters of the equipment for the test.
- Check that the discontinuities that are expected are detected and the measurements are repeated.
- Each procedure requires a pattern.
- Detectabilities: 5~7mm

Requirements

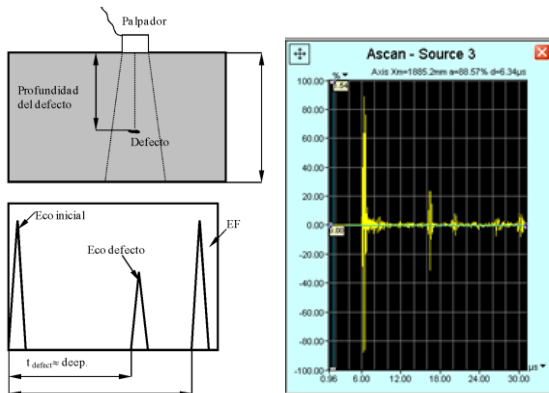
- Pattern material = Part material to be inspected.
- Simulation of discontinuities.



SYSTEMS OF REPRESENTATION

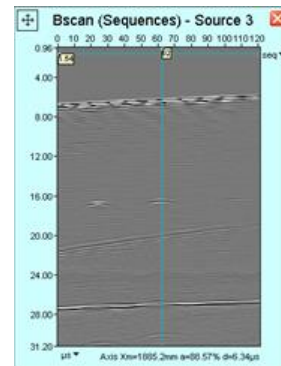
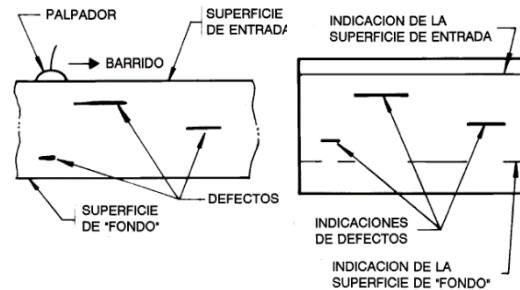
A-scan

- Amplitude of the eco-time (or depth).
- Point representation.
- Manual inspection
- It does not leave a record



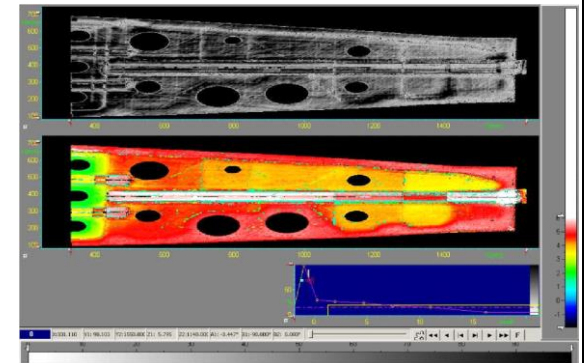
B-scan

- Transversal section



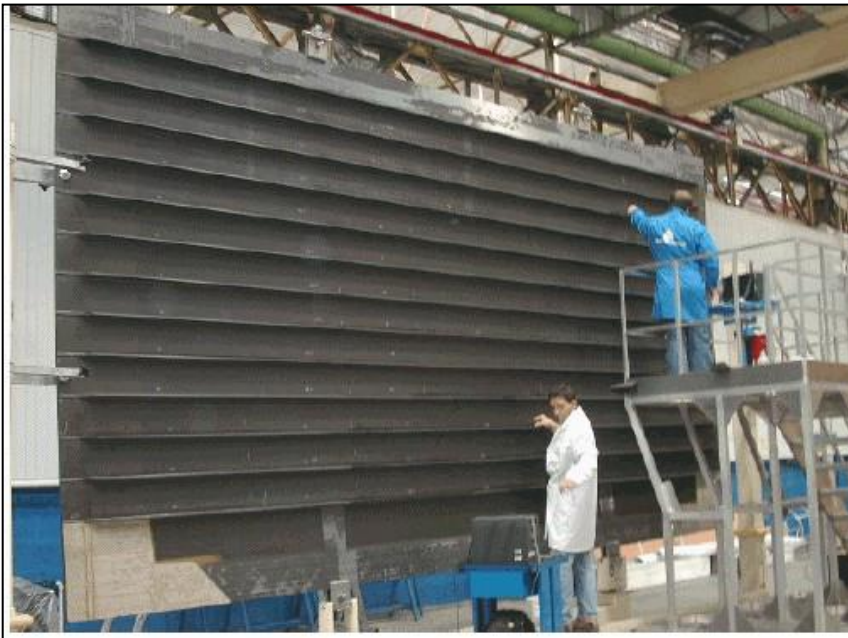
C-scan

- Representation in 2D.

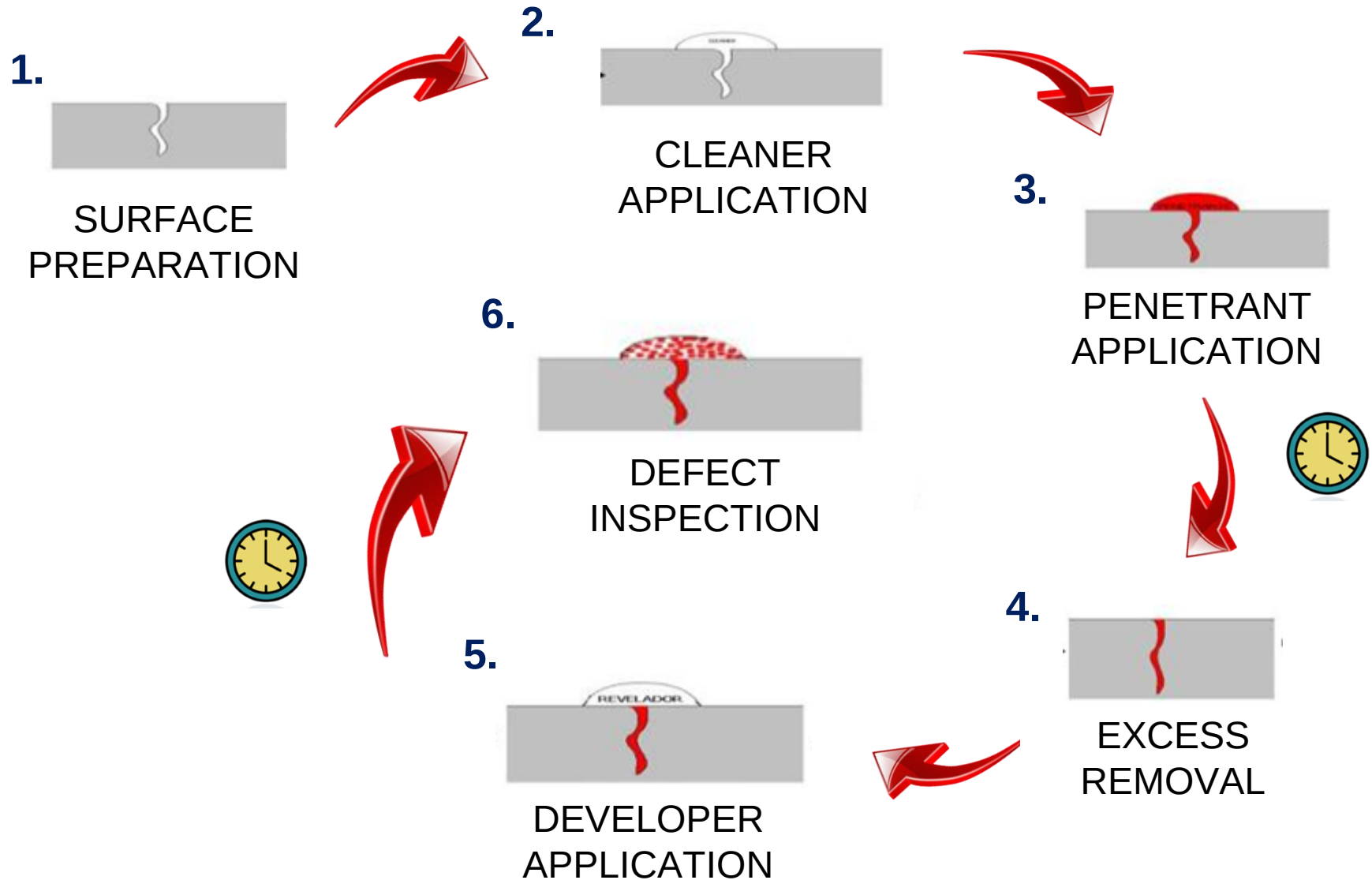


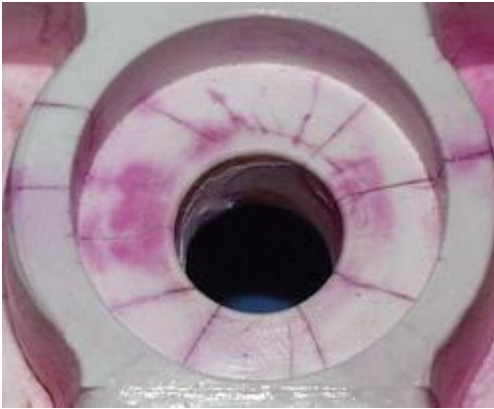
MAIN APPLICATIONS

- Detection of defects, cracks and discontinuities on the surface and below the surface of most materials.
- Thickness measurement

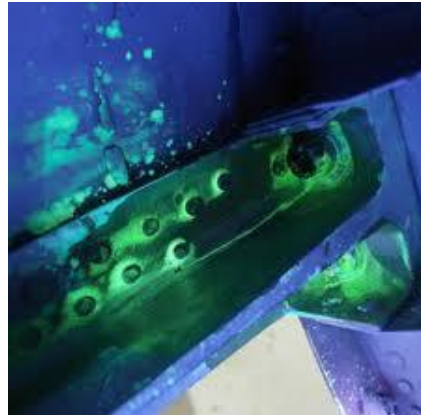


ADVANTAGES	DISADVANTAGES
— Fast, reliable and easy. — Immediate results. — Great precision and sensitivity — The depth of penetration for the detection or measurement of the defect is superior to other methods. — Access is required only on one side of the part or structure to be inspected. — Portable equipment	— Reference patterns. — The surface to be inspected must be accessible. — Surface finish or roughness may interfere with inspection. — Small thickness parts can be difficult to inspect. — Defects oriented parallel to the sound beam can remain undetected.





Visible Dye



Fluorescent



Penetrant application



Method D: Post-
Emulsifiable,
Hydrophilic



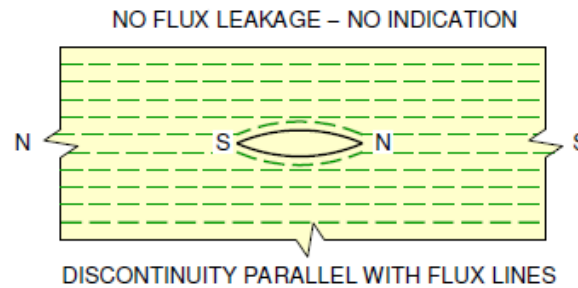
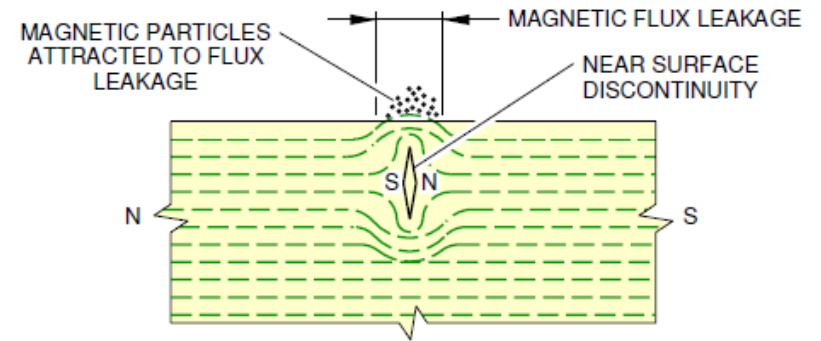
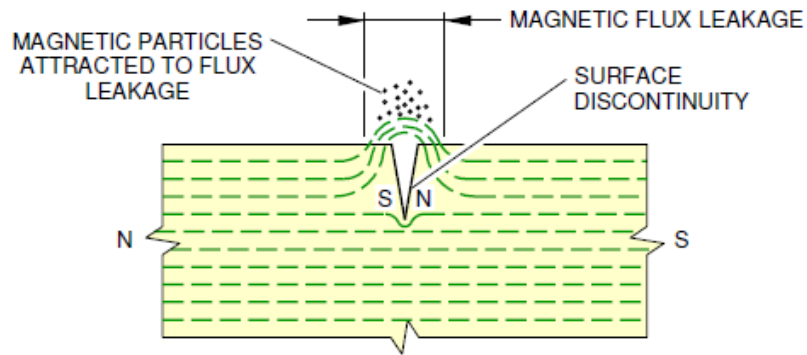
Nonaqueous developer

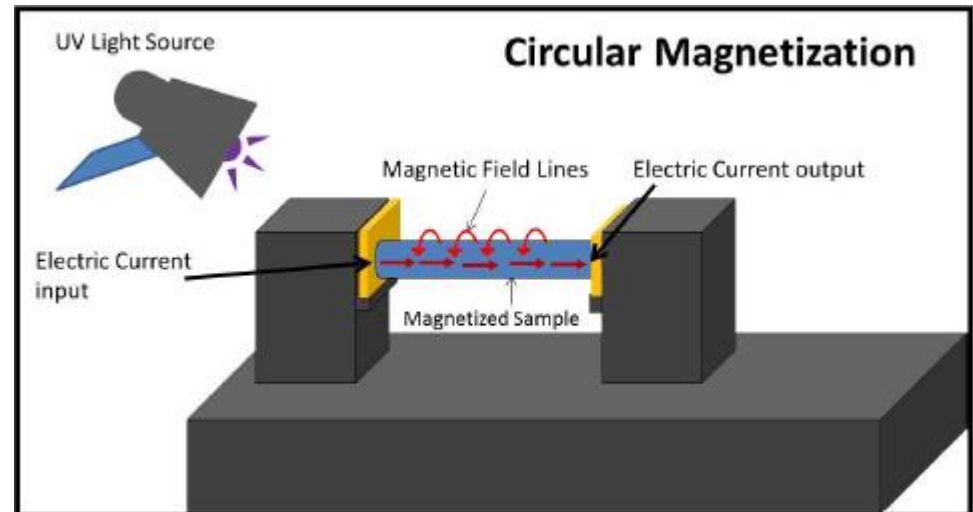
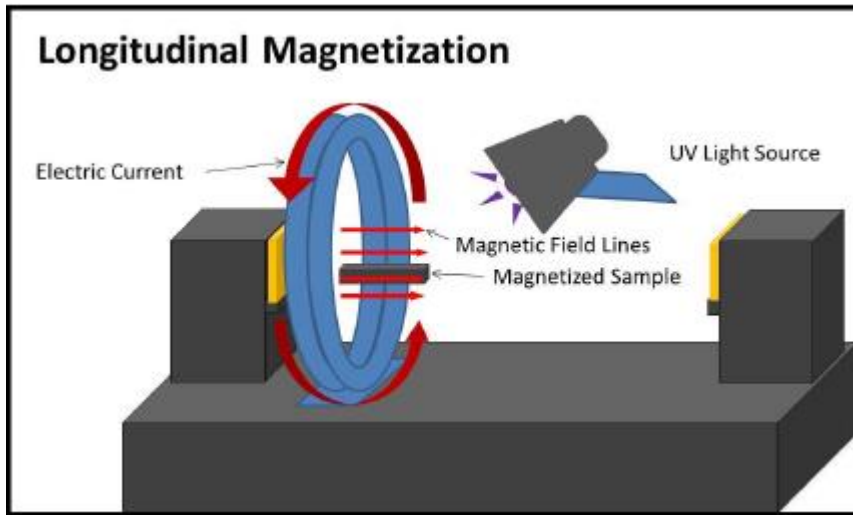


UV mercury vapor lamp

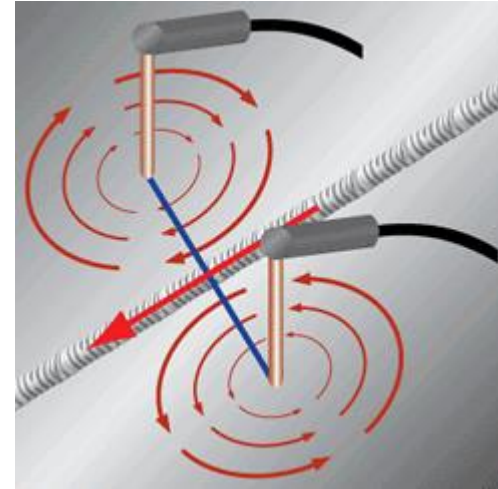
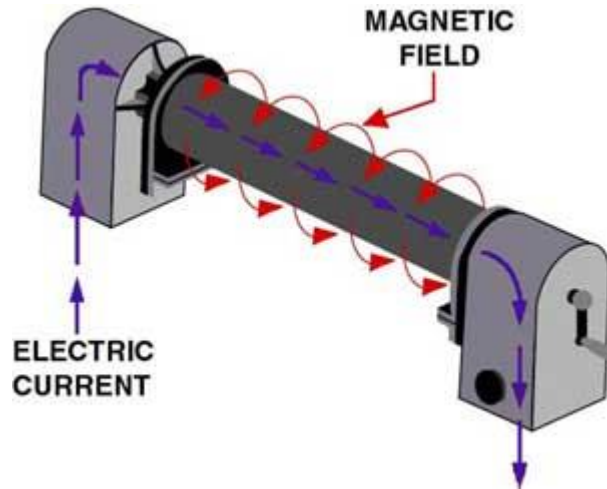
ADVANTAGES	DISADVANTAGES
— Great sensitivity — Few limitations of material — Large surfaces are inspected quickly and at low cost — Very effective for complex shapes. — Indication of direct and visual crack. — The materials used are portable. — Materials and equipment are relatively cheap.	— Only for superficial defects. — Only for non-porous materials (Non-composite). — The previous cleaning process is decisive. — The inspected surface must be directly visible. — The surface finish greatly affects sensitivity. — The sensitivity of the method is very variable. — A subsequent cleaning is required. — It is required to handle toxic / dangerous chemicals. — Materials and equipment are relatively cheap

PHYSICAL PRINCIPLE

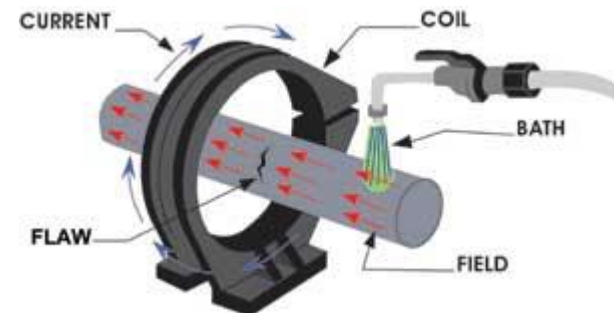
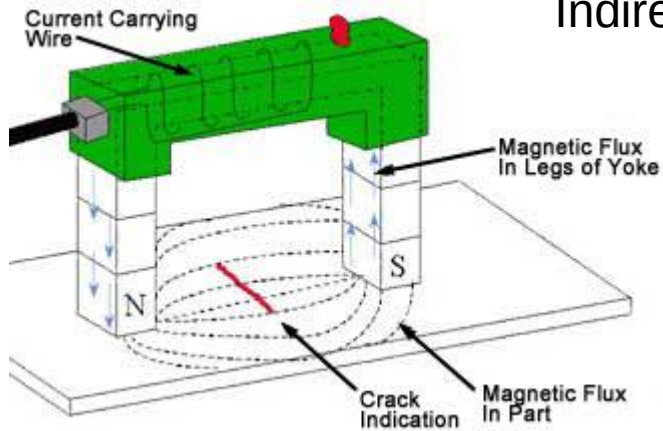




Direct Magnetization



Indirect Magnetization





Visible Dry Particle Inspection



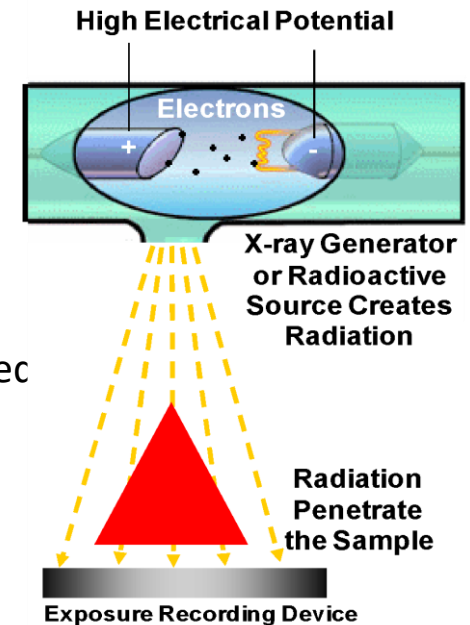
Fluorescent Wet
Suspension Inspection



ADVANTAGES	DISADVANTAGES
— Large surfaces can be inspected quickly and at low cost. — Surface (AC) and subsurface (DC) defects are detected. — The preparation of the surface is not complicated or as critical as LPI. — The indications are produced directly on the surface and form a clear image of the defect. — Equipment costs are relatively low.	— Only ferromagnetic materials. — Very important alignment between magnetic field and defect. — It requires a stable magnetic field. — For large pieces it is necessary to use large currents. — It requires a relatively smooth surface finish. — Non-ferromagnetic surface finishes (paint) negatively affect sensitivity. — Demagnetization and cleaning after inspection

PHYSICAL PRINCIPLE

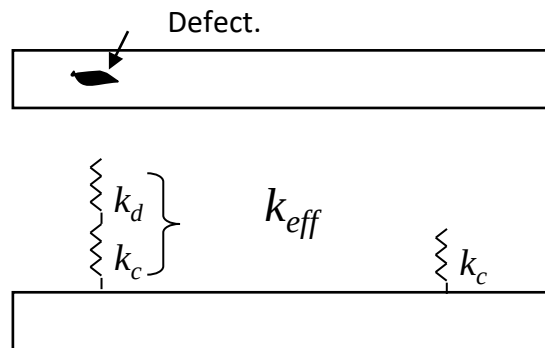
- It is based on the use of high frequency electromagnetic radiation, with the ability to penetrate and pass through materials opaque to visible light, recording images of these materials.
- Representation systems:
 - films
 - radiation sensitive detectors, shaping the image on a screen
- The records are permanent.
- Variables on which it depends:
 - Voltage (kV).
 - Intensity (mA).
 - Exposure time (sec).
 - Nature, density and thickness of the material to be radiographed
 - Distance focus-film / detector.
 - Reinforcing plates (in aviation, only lead plates are used).
 - Type of films or detectors.



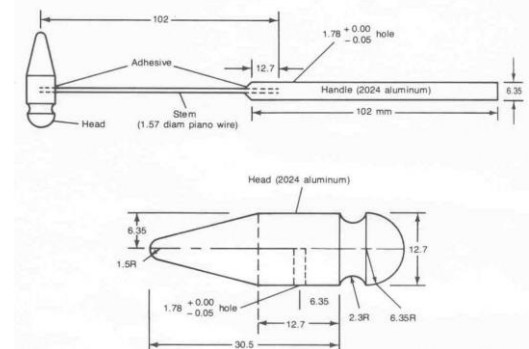
ADVANTAGES	DISADVANTAGES
– All types of materials can be inspected. – Detects surface and subsurface defects . – Ability to examine complex parts and composite material, without disassembling the parts.	– In general, access is required on both sides of the structure. – The orientation of the radiation beam to the non-volumetric defects is critical. – Expensive equipment – Possible radiation hazard for operators

PHYSICAL PRINCIPLE

- It is based on differentiating the sound produced by tapping the structure.

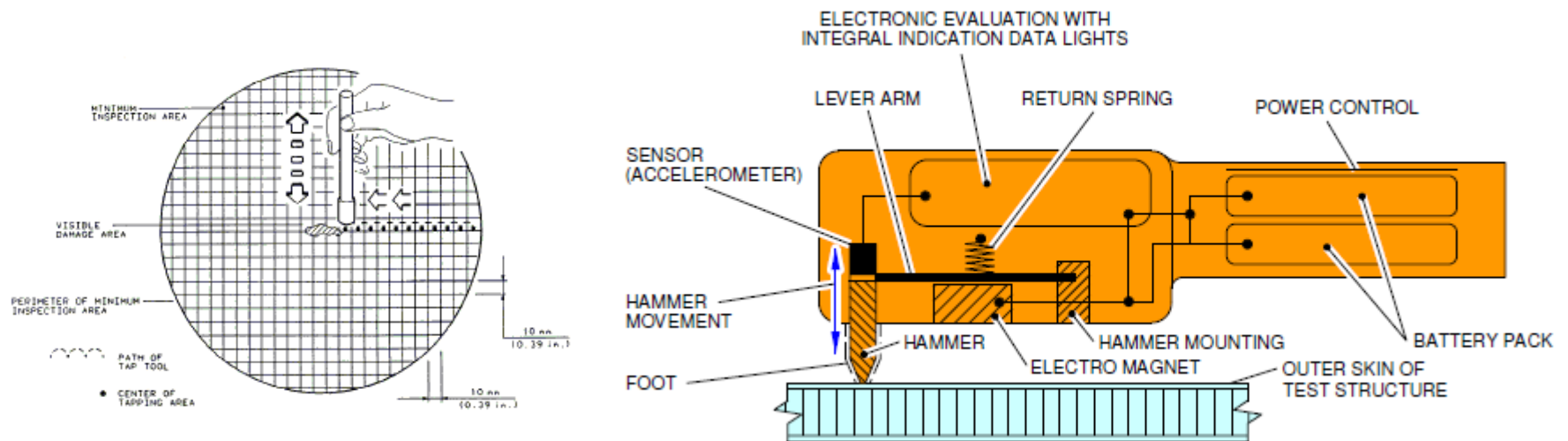


- Different sounds are produced depending on the stiffness:
 - Defective area Low rigidity: More "soft" sound.
 - Non-defective area. High rigidity: More resonant sound.
- Depends a lot on the geometry (thickness, type of structure, etc).
- Very sensitive method.
- Reference test tubes and experience of the inspector.



MAIN APPLICATIONS

- In CFRP, fundamentally in service inspections for the detection of delaminations and debondings, near the surface.
- Applicable for detection of defects greater than 25mm in diameter.
- The inspection can be manual, or with some degree of automation



1. I+D-E-136 Magnetic Particle Inspection Process
2. I+D-E-137 Penetrant Inspection Process
3. I+D-E-139 Crack detection through Eddy Currents
4. AITM6-7002 Radiographic Inspection
5. I+D-E-276 Ultrasonic Testing
6. I+D-E-275 Non-destructive inspection of carbon fiber composite parts
7. I+D-E-294 TapTesting on bonded elements and/or composite structures
8. DOT/FAA/AR-01/95: "Study of the Factors Affecting the Sensitivity of Liquid Penetrant Inspections: Review of Literature Published from 1970 to 1998" January 2002
9. Nondestructive Evaluation (NDE) Capabilities Data Book, Rummel, W.D. and Matzkanin, G. A., NTIAC #DB-95-02, May 1996.
10. NTIAC-TA-00-01: Probability of Detection (POD) for Nondestructive Evaluation (NDE) . August 2001
11. AITM6-0014. Airbus Test Method. Probability of Detection (POD)
12. <http://www.ndt-ed.org/>
13. NDTM A330/340
14. NDTM CN235
15. MSG-3 y su Aplicación en el Desarrollo de los Programas de Mantenimiento. SENASA